

SITE INVESTIGATIONS

DEP 34.11.00.10-Gen.

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DESIGN AND ENGINEERING PRACTICE



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1. INTRODUCTION

1.1 SCOPE

This DEP specifies requirements and gives recommendations for the survey and investigation activities required for obtaining data with respect to the physical site conditions which will affect a construction project.

Assessment of contamination and environmental conditions is excluded from the scope of this DEP since different standards apply to drilling equipment, sampling procedures etc. It may not be assumed that samples taken as part of geotechnical site investigations are suitable for environmental testing. For recommended practices for environmental site assessment refer to SIOF report OP 96-30326.

This DEP is a revision of the DEP of the same number dated March 1985.

1.2 DISTRIBUTION, INTENDED USE AND REGULATORY CONSIDERATIONS

Unless otherwise authorised by SIOF and SIEP, the distribution of this DEP is confined to companies forming part of the Royal Dutch/Shell Group or managed by a Group company, and to Contractors nominated by them (i.e. the distribution code is "C", as defined in DEP 00.00.05.05-Gen.).

This DEP is intended for use in oil refineries, chemical plants, gas plants, supply/marketing installations and, where applicable, oil and gas production facilities.

If national and/or local regulations exist in which some of the requirements may be more stringent than in this DEP the Contractor shall determine by careful scrutiny which of the requirements are the more stringent and which combination of requirements will be acceptable as regards safety, environmental, economic and legal aspects. In all cases the Contractor shall inform the Principal of any deviation from the requirements of this DEP which is considered to be necessary in order to comply with national and/or local regulations. The Principal may then negotiate with the Authorities concerned with the object of obtaining agreement to follow this DEP as closely as possible.

1.3 DEFINITIONS

1.3.1 General definitions

The **Principal** is the party which initiates the project and ultimately pays for its design and construction. The Principal will generally specify the technical requirements. The Principal may also include an agent or consultant, authorised to act for, and on behalf of, the Principal.

The **Contractor** is the party which carries out all or part of the design, engineering, procurement, construction, commissioning or management of a project or operation of a facility. The Principal may sometimes undertake all or part of the duties of the Contractor.

The **Manufacturer/Supplier** is the party which manufactures or supplies equipment and services to perform the duties specified by the Contractor.

The word **shall** indicates a requirement.

The word **should** indicates a recommendation.

1.3.2 Specific definitions

Ground	Soil, rock and fill existing in place prior to the execution of the site investigation or construction project
Survey Contractor	Party appointed by the Principal or the Contractor which carries out one or more of the site investigation activities.

1.4 CROSS-REFERENCES

Where cross-references to other parts of this DEP are made, the referenced section number is shown in brackets. Other documents referenced by this DEP are listed in (13).

2. GENERAL

2.1 PURPOSE

The purpose of a site investigation is to acquire and interpret reliable information for one or more of the following phases of a construction project:

- feasibility;
- site selection;
- basis of design;
- safe and economic design of facilities;
- assessment of hazards associated with the ground;
- location of resources (e.g. sources of sand or aggregates);
- assessment of materials to be excavated or dredged;
- tendering and construction;
- operation and maintenance;
- extension/upgrading;
- de-commissioning;
- assessment of contamination (outside the scope of this DEP).

It is important that those involved in procuring, supervising, interpreting and using data from site investigations work within the reality of how the industry works. In general things are not perfect: techniques are crude; the “representativeness” of samples and repeatability of test results are limited; it may be impossible to sample key strata. Site investigations and the interpretation thereof shall be designed to be robust under these circumstances and to give reliable data despite these constraints.

Because the ground is variable and site investigation information will never be fully representative, in order to reduce the possibilities of misunderstandings and claims, careful consideration is required as to which information is passed to others as “rely upon” and which information is “to be verified” (by the recipient).

2.2 BASIC REQUIREMENTS

This DEP shall be applied as far as relevant to every site investigation.

Supplementary to this DEP, all work shall be carried out in accordance with recognised and accepted theories, methods, codes of practice and standards, and the principles of good engineering practice.

Examples of acceptable standards and codes of practice are given in this DEP.

2.3 UNITS OF MEASUREMENT

SI units shall be used for all surveys and related activities in accordance with DEP 00.00.20.10-Gen.

2.4 FINANCIAL CONSIDERATIONS AND EFFECTS

Site investigations are often carried out at a stage in a project when budgets are restricted because the final investment decision has not been made. There is always pressure to minimise costs and therefore scope. Also, the site layout may not yet be certain.

The extent of the investigation has significant implications for the cost of a project. If only limited information is available to the Contractor, a conservative approach is adopted and costed. Since foundation and site preparation works are the first site works, the Contractor

may not have time to carry out further investigations and optimise the foundation design. For example, piled foundations are often over-designed by a factor of 1.25 or more. This waste, which is ultimately paid for by the client, can be greatly reduced if sufficient site investigation data is available in time.

Additionally, for civil engineering works some 50% of cost and time over-runs are attributable to inadequate information about the ground.

It is therefore important to strike the correct balance between investing (in site investigations) in a project before go-ahead is given, thereby avoiding the risk that boreholes etc. are in the wrong place, and optimising total project cost.

Site investigations should always have the purpose of reducing risk and saving money. They therefore need proper specification and direction.

Mobilisation and provision of access can form a large proportion of site investigation costs, particularly over water, on foreshores, in swamps, on steep terrain and wherever obtaining permits is time-consuming and expensive. Significant savings can be made by coordination: minimising the number of mobilisations, sharing access for other reasons, streamlining permit applications etc.

2.5 COMMUNICATIONS AND CONTINUITY OF STAFFING

The Principal should appoint or nominate a Project Geotechnical Engineer (PGE) at the start of the project, who remains the Principal's focal point for all geotechnical matters for the duration of the contract, thereby promoting continuity. Similarly, the Survey Contractor (and/or main engineering Contractor) shall have continuity in the geotechnical engineer(s) responsible for the relevant parts of the project.

2.6 PROCUREMENT OF SITE INVESTIGATIONS

Advice on how best to procure site investigations, including flow charts which indicate the requisite steps, is provided in Appendix 1.

3. TECHNICAL SPECIFICATION

The Principal shall clearly define the objectives of the site investigation. This shall be in terms of required deliverables (e.g. geotechnical sections and design parameters) and minimum scope of work. The objectives shall include a statement as to which phases of the project will be covered by the data from the investigation.

The Principal shall prepare (or have prepared) a specification and scope of works which will achieve the objectives.

Minimum general requirements for site investigations for normal buildings and structures are those set out in this DEP, in BS 5930 and in BS 1377. Additional specific requirements given in the relevant DEPs and external standards shall be followed where applicable. Examples of such external standards are:

- BS 6031, Code of practice for earthworks;
- BS 6349, Code of practice of maritime structures;
- BS 7777, Code of practice for storage tanks for low-temperature service;
- BS 8002, Code of practice for earth retaining structures;
- BS 8004, Code of practice for foundations;
- BS 8005, Guide to design and construction of outfalls;
- BS 8010, Code of practice for pipelines;
- BS 8081, Code of practice for ground anchorages.

ASTM volumes 04.08 and 04.09 contain the ASTM standards. These include some tests and procedures not covered by British Standards.

A single set of standards shall be used, preferably British Standards (ASTM standards may be used in countries where this is normal practice, and for subjects not covered by British Standards). This helps to reduce inconsistencies, overlaps and gaps.

The use of site investigation measuring equipment shall be in accordance with ISO 10012-1.

The Principal shall ensure the preparation and implementation of a project plan, preferably in accordance with ISO 9004-5. The outline shall normally be done by the Principal, with the details being done by the Survey Contractor. The project plan shall include:

- a flexible approach that allows adjustment of the site investigation process, as information becomes available;
- clear lines of communication;
- a continuous chain of responsibility for the site investigation process and for decisions with respect to site conditions;
- details of equipment, survey methods, qualifications of personnel;
- programme.

The Survey Contractor shall keep the relevant parts of the specification, external standards, project plan, work instructions or method statements available for inspection at or near the place of the activities (for example at the site office, the laboratory and the Survey Contractor's office).

Since the Principal will normally supply the results of the factual site investigation to the main engineering Contractor as "rely-upon information", any data items not considered reliable shall be so marked. Relevant interpretative information shall normally be provided "for information only" with a contractual obligation for the Contractor to verify it and to take responsibility for the interpretation used.

4. SITE INVESTIGATION CRITERIA

4.1 GENERAL

The type and scope of a site investigation depend on the purpose for which the investigation is required and on the circumstances under which it is to be carried out. The various types and scopes of site investigations are described in (4.2).

The tools or techniques available for a site investigation consist of:

- desk studies;
- field surveys, such as boreholes, CPTs, in-situ tests, visual inspection, measurements and sample collection;
- laboratory testing of samples collected during a field survey;
- data processing, integration and interpretation techniques for reporting purposes.

Consideration shall be given to:

- obtaining adequate site investigation data at an early stage in the project, as uncertainty about site conditions will be reflected in bid prices by Contractors;
- providing specialist supervision of the Survey Contractor's field survey activities, in order to help in ensuring that the objectives are met (e.g. by adjusting the investigation plan and detecting any deficiencies);
- preserving samples of ground or groundwater and retaining specific monitoring instruments for ongoing or future use.

4.2 PHASED INVESTIGATION

4.2.1 Phasing

Stages in a project when a site investigation may be carried out are listed in (2.1). For each project stage a certain amount of detail is required. If the investigations for previous stages provide this level of detail, no further investigations are required for the current project stage (unless required for verification). If not, additional investigations are required.

The time required for a ground investigation is typically in the order of 3 to 6 months plus contract preparation, tendering, tender evaluation etc. Mobilisation also forms a substantial proportion of the costs. There are therefore benefits in carrying out a small number of field investigations, each of which are sufficient for several project stages, rather than a separate field investigation for each stage. More information on this is provided in Appendix 1.

4.2.2 Desk study and walkover survey

Consultation of available data is often sufficient broadly to categorise a site and give a general idea of the required foundations and probable geotechnical problems. It is normally carried out at the start of a project. Data typically available are:

- topographical maps;
- geological maps;
- geological records;
- engineering geological maps;
- previous ground investigation reports;
- construction records (e.g. pile driving and test pile records);
- seismic risk maps + earthquake data;
- soil survey maps (for agriculture);
- well supply memoirs and records;
- mining records;
- water authority records;
- local government records;
- historical data (old maps, previous use, changed landforms);
- utilities crossing the site;
- aerial and other photographs;
- meteorological records;
- flood records;
- newspaper and technical journal records of other hazards such as landslides, mudslides and or volcanoes;
- Internet sites.

Typical sources of data are:

- local authorities;
- geological survey (government body);
- local museums;
- client (if site is being extended);

- utilities companies;
- government mining office;
- mining companies;
- staff who have been in the area;
- local residents;
- building regulations;
- site visit/walkover survey;
- old newspapers (newspaper record departments).

A report sheet for a walkover survey is provided in Appendix 1.

4.2.3 Preliminary site investigation

A preliminary investigation is normally carried out in the initial stages of the contract (feasibility study, scouting study or Basis of Design). It typically comprises a limited number of boreholes, CPT tests and/or other exploratory holes spaced widely across the site. The objective is to give a general indication of ground conditions and to identify any potential problems. If structure locations are known, the boreholes/CPTs may be located to reflect this, otherwise a grid is normally applied. Grid spacing is typically between 50 m and 500 m depending on how well the project and plant locations are defined and the probability of the project going ahead. Other surveys which may be carried out at this stage include topography, bathymetry, seismic hazard or other geo-hazard, especially when little information is available from other sources.

The preliminary investigation is normally initiated by, and carried out on behalf of, the Principal.

4.2.4 Main site investigation

The main site investigation for the Basic Design and Engineering Package (BDEP) stage or for a tender design is usually an additional survey. It is normally initiated by, and carried out on behalf of, the Principal after the layout of the scheme is known (albeit not necessarily in detail). The objective is to provide sufficient information to permit tender designs and cost estimates of sufficient accuracy for the final investment decision and for Engineering, Procurement and Construction (EPC) bids. It is not generally sufficient for detailed design purposes for an EPC contract. Where the investment decision is fairly certain and the plant layout is well defined, it is often appropriate to carry out a full detailed investigation at this stage, as this will give a time and cost saving for the project.

It typically comprises investigatory holes at approximately 50 m to 100 m spacing in plant areas and at the centre and 4 points around the circumference of tanks. Topographic, bathymetric geophysical, seismic hazard and geo-hazard surveys are typically included at this stage if they have not already been performed.

4.2.5 Detailed site investigation

The detailed site investigation (investigation for detailed design) has the objective of producing all data required for detailed design. Normally, the layout of plant and equipment is known at this time.

In an EPC contract, the Contractor should be responsible for, or should be consulted throughout, the detailed investigation (this is important in view of his ultimate responsibility for the construction project).

In an EPC contract the Contractor will normally be required to assume full responsibility for all site investigation data and interpreted results. If the Contractor has doubts about any of the data or considers it inadequate, he should carry out a verification investigation and/or additional investigation.

4.2.6 Verification investigation

If a Contractor uses the results of available/existing investigations carried out on behalf of the Principal or a third party, and agrees to accept the recommendations of such investigations within the terms of his contract, the Contractor shall carry out a 'verification survey' (unless it is unnecessary e.g. because the information provided is specifically detailed investigation information for the work, is fully trustworthy and gives full coverage).

The objective of a 'verification survey' is to verify the data established by earlier investigations and to confirm the basis on which the relevant recommendations have been made. The scope should therefore be limited to what is considered necessary in this respect.

Subject to the results of the 'verification survey' being in agreement with the results of previous investigations, within acceptable limits, the Contractor shall take the responsibility for all survey data and related reports and recommendations used for design and construction.

A verification investigation may also be required during the construction phase. If it becomes evident that survey data from investigation, analysis or observation appear to contradict the corresponding equivalent data obtained from other or earlier surveys, a 'clarification survey' shall be performed.

The scope of such a clarification should be limited to the extent necessary to determine the cause of such contradictions and reconcile them.

Verification survey results shall not absolve the Contractor from responsibility for all geotechnical data and interpretation he uses.

4.2.7 Monitoring investigations

Construction monitoring may include activities such as checks on fill compaction using CPT or SPT testing, ground water levels, logging of ground conditions in foundation excavations etc. Recommendations on monitoring programmes should be part of interpretative survey reports and/or method statements. This is covered by DEP 34.11.00.11-Gen, and is therefore not covered in this DEP except for the investigation techniques.

4.3 APPLICATION OF SITE INVESTIGATIONS / SURVEYS FOR PROJECTS

4.3.1 Extent of investigation required

The extent of information required depends on the nature and complexity of the works to be constructed, the nature and complexity of the ground and the acceptable risk of failure (related to direct effects and economic damage that can result). Guidance on scope is provided in Appendix 1.

The investigation will include some or all of the survey types described in sections 5 to 12 and will be phased as described in section 4.2. The level of detail shall be appropriate to satisfy the objectives of the investigation.

4.3.2 Major projects

For major projects a staged investigation will normally be required.

4.3.3 Minor projects

For small projects and small extensions to existing facilities and structures, available survey (or design) data may be used if it is sufficient. Normally, such data should be verified by means of a 'verification investigation'. Depending on the scope of the works this may vary from a detailed investigation to inspection of the exposed base of an excavation for a shallow foundation, supplemented by manual testing.

4.3.4 Projects with an "over-water" component

If there is surface water in the area to be investigated the cost of the project site investigation will be substantially greater than that of a similar dry-land project. Mobilisation, moving, access and down-time costs are generally higher. The inter-tidal zone causes particular problems because it can be difficult to reach with either onshore or marine equipment. Consideration shall be given to factors such as:

- it may be cost-effective to minimise the number of investigation phases of a nearshore site investigation because of the limited availability and high mobilisation cost of adequate floating craft and special field survey apparatus (refer to Appendix 2);
- progress is often closely dependent on tides, currents, weather conditions and safety aspects;
- feasibility and timing of a geophysical survey is particularly important with respect to the scope of the geotechnical survey. There are potential cost and planning benefits from co-ordinating a nearshore geophysical survey with a bathymetric survey, as they can usually be carried out by the same boat at the same time.

4.3.5 Procurement of site investigation/survey contracts

Competitive tendering should be applied.

It is important to pre-qualify tenderers on the basis of the scope of the work and their local expertise. Prequalification procedures should include submissions covering experience, proposed staff, equipment, work methods etc. by the Contractor (for known Contractors these submissions can be left until the tender), and consultation with others who have used the Contractor.

Interpretative reporting responsibilities should only be awarded to firms who have competent staff available to do the work, and who can provide assurances that these staff will be available when planned.

A good geotechnical interpretative report can in general only be written by someone who has been on site and seen the ground first hand. Descriptions by others are usually subjective and rarely convey full details of the site.

4.3.6 Procurement of geotechnical specialists

Specialist geotechnical expertise may have to be procured for:

- carrying out desk studies;
- supervision of field work;
- review of Survey Contractor submissions;
- design review.

Such expertise is normally procured from geotechnical consultants or other contractors. In the selection of specialists, attention shall be paid to the individual's experience and knowledge, both of local ground conditions and foundations, and of the type of structures to be constructed. Back-up that the individual can receive from his company is also important.

Specialist expertise should be procured on the basis of technical merit rather than price. The cost of the individual expert is normally a small part of the geotechnical costs on site, and the correct expertise is likely to be more cost-effective.

Close and active co-operation between individual specialists is required at every stage of the project if an optimal result is to be obtained and unnecessary delays are to be avoided in comment rounds.

For specialist subjects or ground conditions, experts such as university professors may be consulted. In selecting such experts, attention shall also be paid to their availability and their ability to deliver results in an acceptable format and timescale.

4.4 REPORTING

The result of a site investigation is a usually report or a series of reports. Several years after the investigations are carried out the people who supervised the investigations and wrote the reports may no longer be available. It is therefore essential that the reports are of a high standard, fully describing the site conditions and highlighting any places where the data may be uncertain, incorrect or possibly misleading (the latter may occur for example if it was impossible to access the worst part of site, because samples of weaker materials were too weak to test, because it was a particularly mild winter when wave tracking was done etc.). External standards usually provide detailed specifications for the presentation of factual data; in general, BS 5930 and BS 1377 shall be followed.

The essential reporting requirements shall be specified and shall include:

- a) the purpose of the report;
- b) summary descriptions of work carried out, adopted practice, procedures or method statements, with references to external standards where appropriate;
- c) a list of previous investigations carried out at the site and of sources of information consulted;
- d) results;
- e) interpretation of results where required;
- f) hard copies of draft and final report;
- g) electronic copy of final report;
- h) comments on circumstances that may lead to defective, irrelevant, insufficient or inaccurate data.

If the reliability of the conclusions and recommendations may be inhibited by insufficient tangible evidence or experience, the Survey Contractor shall identify where additional studies are required in order to provide the required degree of confidence.

Particular consideration shall be given to specification of the scope (number and type of analyses, detailed advice, recommendations etc.) and the required level of detail and sophistication of interpretative analysis. This promotes:

- (1) reduction of bid uncertainties incorporated in proposals by potential Survey Contractors;
- (2) ready evaluation of proposals of Survey Contractors; and
- (3) adequate checks of completed activities.

The basic types of site investigation reports are:

- a) desk study report;
- b) daily field (or laboratory) report (typically hand written logs etc.);
- c) weekly field (or laboratory) report (normally typed);
- d) final field report (not normally required if weekly reports have been submitted);
- e) draft and final factual report;
- f) draft and final interpretative report.

The need for a particular type of report depends on the survey requirements and the nature of the project. For example, daily reports on a geotechnical survey are useful where preliminary information is urgently required for design or for determining the magnitude of further site investigation.

Presentation of field data consists of operational information and preliminary factual data.

Interpretative reporting shall include appropriate evaluation of the factual data. In particular, adverse data shall be considered carefully in order to determine whether they are

misleading or represent a real phenomenon that must be accounted for. The presentation of interpretative information may include sketches, photographs, plans and drawings, cross-sections, contour plans, tabular and graphical data. Complex projects may require video presentations, 3-D computer models and Geographical Information Systems.

4.5 HEALTH, SAFETY AND ENVIRONMENT

4.5.1 General

All work shall be carried out in a safe manner. The relevant health, safety and environmental warnings in the applicable external standards shall be heeded.

It should be realised that site investigation equipment and procedures are by their nature higher risk than most, as work takes place in unprepared locations, with numerous lifts of (heavy) equipment, in the presence of slippery drilling fluids etc. Hence, safety and safe-thinking must be a priority if accidents are to be avoided.

At the tender stage each tenderer shall submit an HSE plan, focused on the work tendered for. This shall include an HSE Manager who shall be mobilised to site with the first personnel. Potential Contractors' HSE records shall be included in prequalification/tender evaluation. An unsatisfactory HSE record shall be grounds to disqualify a Contractor.

4.5.2 Existing services and facilities

In the event of site investigations within or close to locations having existing services and facilities, adequate precautions shall be taken to protect such services and facilities from damage or interference during the execution of the work.

Before commencing any site investigation which will require excavations, borings or penetration testing of the ground, the nature and location of all existing services and facilities on the site and adjacent areas shall be ascertained. Typical sources of such information are utilities companies, site drawings, maintenance departments and visual observations (e.g. location of manholes). This preliminary investigation shall cover all such services and facilities aboveground and underground. The use of a cable detector tool calibrated against known underground services can be useful. Utilities and pipelines are often present even though not marked on a drawing (or wrongly marked). If there is any reason to suspect that some such item may be present (or insufficient justification to prove that nothing is present), manual excavation to typically 1.5 m depth shall be carried out prior to commencing the exploratory hole.

The Principal shall be notified should these investigations reveal any objects of which there is no previous record. A description, dimensions, levels related to reference datum points and other appropriate details of all such objects shall be recorded accurately and supplied to the Principal.

For services that can be isolated during the course of the survey, a certificate or permit to work shall first be obtained from the Principal or other responsible body, confirming that such services have been isolated and that it is safe to proceed with the site investigation. For services and facilities which need to remain in operation, the Principal shall specify exactly what these are and what requirements shall be met whilst the investigations are being carried out. The Contractor or other party carrying out the investigations shall be responsible for strict adherence to all safety regulations in force at the location concerned.

In the event that a service or facility is cut or otherwise damaged accidentally, it shall be reported to the Principal immediately and corrected as directed.

5. SITE SELECTION SURVEY

5.1 GENERAL

The selection of a construction site for a refinery, gas plant, chemical plant, terminal or harbour shall be based on economic and technical considerations, taking into account the development plans of the local situation, and the layout and design criteria for the construction project.

The following aspects shall be considered in a site selection survey:

- 1) Previous site uses:
 - effects of previous land use (obstructions, disturbance, contamination, holes in impervious layers, mine workings beneath site that could subside, etc.).
- 2) Topography:
 - available area (versus area required, including the requirements for future extensions);
 - variation in ground levels;
 - ground level relative to open water level;
 - obstacles, roads, rivers, coastlines, vegetation, etc.;
 - earth moving requirements.
- 3) Geology/Geotechnics:
 - ground conditions and geology;
 - seismic risk (if relevant);
 - presence of faults;
 - presence of faults/cavities/karst features;
 - bearing capacity, settlements and stability (short and long term);
 - borrow and quarry resources;
 - chemical properties/stability, durability, degree of weathering of substratum/subsoil.
- 4) Geo-hydrology:
 - groundwater level and variations therein;
 - groundwater flow/permeability;
 - chemical properties of groundwater, and its possible aggressiveness;
 - surface run-off;
 - drainage requirements (surface and subsurface);
 - influence of the proposed project on the above;
 - risk of groundwater contamination from the project, and sensitivity.
- 5) Meteorology:
 - ambient temperature of air and water;
 - precipitation;
 - wind (cyclone/typhoon conditions, if relevant);
 - pollution/contamination;
 - vegetation;
 - observations with respect to cooling water intake/outfall.

- 6) Oceanography:
 - waves;
 - currents;
 - tides;
 - (sea) ice;
 - morphology;
 - character of rivers (including flooding, siltation, debris, etc.);
 - tsunamis.
- 7) Miscellaneous:
 - ecology;
 - archaeology and ancient monuments;
 - environmental sensitivity of the area (physical and political);
 - upstream dams and their integrity;
 - native heritage/title or other land ownership disputes, whilst not part of the investigation, should be brought to the attention of the Principal if encountered.

5.2 INTERPRETATION AND REPORTING

Requirements for the interpretation and reporting of a site selection survey shall be specified by the Principal and shall include the following as a minimum:

- a) an appraisal of the physical conditions at the site;
- b) correlation and integration of available data;
- c) recommendations for further detailed investigations, where applicable;
- d) a discussion on possible benefits, drawbacks and limitations of the site(s) for the proposed project, including future extensions;
- e) conclusions about site selection.

Incorrect conclusions from the investigations are more often arrived at through forced interpretation of the data in favour of the development rather than through lack of ability to evaluate those data correctly. It is important to ensure, therefore, that no factor is assumed to be favourable until that assumption is supported by data. Similarly, unreasonably pessimistic assumptions about unknowns may unreasonably prejudice the calculated viability of a proposed project. If in doubt, reporting should follow a 3-point distribution method: (a) worst credible value, (b) best credible value and (c) best guess/recommended value.

6. GEODETIC SURVEY

6.1 OBJECTIVE

The objective of a geodetic survey for a preliminary investigation is to establish the project location and site investigation locations digitally and on a topographic or bathymetric map using a defined (local) co-ordinate system and level datum with a defined correlation to a recognised, (inter)national projection system and geodetic datum.

The objective of a geodetic survey for more detailed investigations is to provide topographic, bathymetric and location information to the required accuracy for the relevant stage of the project. Location information includes the levels and locations of all exploration points of the various surveys.

6.2 FIELD SURVEY

6.2.1 Objectives

The objectives of the survey vary with the stage of the project. Typically the following are required:

6.2.1.1 Scouting study

A topographic or bathymetric map to a scale of between 1:50 000 and 1:10 000; and reference points for triangulation for topographic and bathymetric surveys. If no map is available, then a rough survey and sketch map shall be made of the area showing the plan and governing elevations of watercourses, and the situation and character of important features such as woods, swamps, pastures, roads, railways, canals, major structures and buildings. Aerial photography or remote sensing data may support this work.

6.2.1.2 Feasibility Study once site has been selected

When the site of a project has been fixed as a result of the preliminary investigations, a project location map should be prepared and based on an accurate field survey. The scale of the map should be such as to provide a 0.5 m contour interval within the plant area and 1.0 m interval in the immediate surroundings. In steeply sloping terrain these contour intervals may be doubled. Outside this area, a larger contour interval may be used, particularly in rugged country or over water. For projects of major importance, an accurate field survey shall include the following:

- 1) A triangulation system laid out for the control of later on-site construction surveys. This shall include a carefully surveyed base line located centrally, and a limited number of documented permanent monument points, which shall be placed strategically and accurately related to the plant and (inter)national grid system, to Project Reference Level, to the datum normally used in maps of the area and, for coastal and river edge sites, to Chart Datum (CD) and Lowest Astronomical Tide (LAT).
- 2) A co-ordinate system as a control for the location of exploration points (e.g. boreholes). If a tie-in to an existing co-ordinate system is not practical, a local co-ordinate system shall be established before the commencement of any drilling or other subsurface investigations.
- 3) Survey maps shall clearly indicate which co-ordinate system has been used and shall include correlations between the local and (inter)national co-ordinate system and height datum.

6.2.1.3 Further Project Stages

Once the site survey has been done, additional geodetic survey is only required:

- if more detail is required locally;

- if an additional area has to be surveyed (change in layout, extension, access routes, etc.);
- for a confirmation survey;
- for construction.

6.3 INTERPRETATION AND REPORTING

6.3.1 General

The results of a geodetic survey do not normally require interpretative analysis (except for drawing contours).

Factors affecting the accuracy and reliability of the results shall be reported, e.g.:

- type(s) and accuracy of survey method, including the use of correction factors;
- proximity and accuracy of control points;
- atmospheric and other environmental conditions;
- movements of the location reference point at the time of the survey, for example if the location reference point is on a vessel;
- the nature of the (underwater) ground surface, in particular where this consists of soft ground. For example, the water pressure device mounted on an underwater frame can sink into the ground surface, thus affecting the recorded water depth. Similarly, an echo sounder may be affected by insufficient acoustic contrast between water and soft ground;
- correction factors for salinity, temperature and barometric pressure can be important for water pressure devices and echo sounders;
- inaccuracies in associated measurements, for example in the measured length of the drill pipe in the case of the level of a sample in a borehole;
- offset of the test or sample location from the location reference point, for example in the case of an inclined drill pipe.

Significant cost savings are possible by combining marine bathymetric surveys and geophysical work, as the equipment for both can be towed at the same time by the one boat.

7. GEOLOGICAL SURVEY

7.1 OBJECTIVE

The objectives of a geological survey can be summarised as follows:

- 1) To provide an understanding of the geology, geomorphology and geological history of the project area, that is of the processes from which the present geological situation at and around the site has been developed. In particular, it is important to determine the following:
 - the general stratigraphy and the distribution of the major ground units, and the boundaries and the form and uniformity of the interface between the major units;
 - any processes that are still active or potentially active: these could include erosion and/or deposition, consolidation or heave, slope creep, landsliding, fault movement, subsidence and volcanic activity;
 - the possible effects of the active processes on the engineering works, both during construction and when in service;
 - possible changes to the effects of active or previously active processes, or to the existing geological condition (e.g. change to the existing stress or hydrological condition) which could arise from the engineering works from the proposed project or from other proposed or existing works (e.g. by third parties) in the area, and whether any remedial action will be necessary;
 - the seismic character of the area;
 - groundwater levels, heads, whether groundwater level is static, whether it is at a natural level or held at a level artificially (e.g. because of water abstraction, poldering, etc.). If it is not natural, could it change because of a change in pumping regime;
 - the direction of flow of groundwater and the local drainage regime;
 - the geomorphology of the project area in terms of the regional stratigraphy, structure and geological history.
- 2) To form a logical basis for the investigation of sources of fill and construction materials.
- 3) To draw attention to important geological features that may represent hazards, constraints on layout, or opportunities for savings for the construction, operation and decommissioning of the project at or close to the site, but not actually exposed or readily recognisable at the site.

Appendix 3 presents a list of potential ground hazards.

7.2 GEOLOGICAL FIELD SURVEY

As for other surveys, the objectives, scope and deliverables shall be specified by the Principal and shall include the following:

- area of interest, as this will generally be larger than area of site itself;
- scope of search of public and proprietary data bases;
- on-site data recording, mapping;
- aerial photography or satellite imagery;
- method of presenting results.

A search of public and proprietary data bases usually considers relevant literature, topographical and geological maps, aerial photography and satellite imagery, seismic events, old maps, descriptions or photographs, earlier site investigation data etc.

Information recorded during engineering geological mapping shall normally include:

- material and/or mass characteristics of ground observed in natural or artificial exposures, for example the orientation and frequency of bedding and jointing discontinuities and weathering profiles;
- structural information on faulting and folding;
- geomorphological features, such as topography, vegetation and drainage patterns;
- man-made features, for example cut and/or fill features, performance of foundations and retaining structures.

7.3 INTERPRETATION AND REPORTING

7.3.1 General geology

The report shall cover the items referred to in 7.1 and 7.2.

7.3.2 Engineering geology

Interpretative analysis shall consider and report on how relevant geological features may affect design, construction or performance of the structures, such as:

- foundation type: safety against sliding, settlement, internal or external erosion;
- formations to be excavated: ease of excavation by various types of earth-moving equipment, whether blasting is required, stability of slopes, support requirements, dewatering, etc.;
- sources of construction materials for use as rock fill, aggregates or fill material, including observations on the workability of the deposits concerned; suitability of excavated ground.

Consideration shall also be given to:

- susceptibility of the ground to possible accelerated natural weathering or other processes brought about by the construction project itself;
- special ground conditions, such as listed in Appendix 3;
- groundwater, for example surface flows, springs, groundwater table, chemical composition.

Useful tools for interpretative analysis and reporting include:

- conventional annotated site plans, including descriptions, symbols and/or classification zoning;
- aerial photographs (marked up with key information shown);
- geological map(s);
- specific data projections, such as stereonet projection of rock discontinuities;
- computer-based data base and/or Geographical Information System (GIS) presentations;
- computer ground models (2D and 3D).

7.3.3 Seismicity and seismic hazard assessment

Seismic hazard analysis can be viewed as a specialist part of the geological survey. A preliminary assessment can be made by selection of seismicity characteristics from regional or nation-wide zoning maps and regulations using appropriate typical ground profiles. From these, design accelerations and response spectra are deduced. For sites where seismic loads may be critical and for structures in seismic areas where no regional or nation-wide zoning maps are available a more detailed seismic hazard analysis is required.

A common approach is as follows:

- initial analysis as above;

- geological desk/field study to locate (potentially active) faults relative to the site;
- extraction of frequency and amplitude characteristics of earthquakes from an up-to-date data base and as interpreted from "non-scientific records" (e.g. historical descriptions of earthquakes in the area) where available;
- estimation of design earthquake from recurrence plot for selected reference return period(s) where feasible, or by worst case scenario;
- estimation of design epicentre distance and rockhead acceleration, on the basis of geological information (possibly including field survey for fault identification), data base information and selected attenuation relationships;
- determination of site response spectra for the required range of ground conditions for operating base earthquake (OBE), for which plant is undamaged, and safe shutdown earthquake (SSE), for which operations can safely be shut down although damage will have been sustained;
- hazard assessment for the proposed construction project.

The seismic hazard assessment shall also consider the effects of phenomena associated with earthquakes, e.g.:

- tsunamis;
- liquefaction;
- mud slides/flows;
- rockfalls;
- etc.

Seismic hazard assessment should also refer to ASCE (1997) "Guidelines for Seismic Evaluation and Design of Petrochemical Facilities", and should comply with the recommendations contained therein.

8. GEOPHYSICAL SURVEY

8.1 OBJECTIVE

The primary objectives of an engineering geophysical survey (often referred to as “seismic survey”) are one or more of the following:

- data acquisition for geological purposes, such as establishing the general stratigraphy between boreholes/CPTs etc., weathering profiles, erosional or structural features (buried channels, faults, dykes etc.);
- resource assessment, such as location and nature of aquifers, sand and gravel deposits, rock for aggregates;
- detection of voids and buried objects, such as mine workings, natural cavities, old foundations, buried tanks;
- detection of environmental hazards, such as hydrocarbon pollution, mine tailings, etc.

In order for the objectives to reflect what can realistically be produced, they should be discussed with potential contractors.

Geophysical surveys should be calibrated against intrusive investigations such as boreholes or CPTs.

8.2 FIELD SURVEY

There are many different geophysical techniques, each based on different theoretical principles, such as seismic velocity and electrical resistivity. They produce different sets of information related to the characteristics of the ground. It is important to use the right equipment, correctly calibrated and tuned to the appropriate water depth. UFG (1992) presents guidance on geophysical techniques.

The common field survey techniques for use on land are:

- seismic refraction for geological purposes, in particular rock excavation assessment. Depth and resolution have few absolute limits, but depend on equipment used, number of sampling points, (i.e. amount spent) etc.;
- surface and downhole electrical resistivity for assessment of corrosion potential and cathodic protection;
- ground probing radar for shallow contamination detection and for detection of (shallow) voids, services etc.;
- cross-hole tomography (downhole, between boreholes) to detect structure, faults, voids;
- downhole shear wave velocity in boreholes (sometimes by Seismic Cone Penetration Test) for earthquake resistant design and soil liquefaction assessment and for design of dynamically loaded foundations.

The common techniques for use over water are:

- echo sounding for determination of bathymetry;
- side-scan sonar for interpretation of seabed features and detection of surface objects;
- seismic reflection for geological purposes, in particular stratigraphic reconnaissance and detection of shallow gas;
- magnetometer profiling for detection of ferrous objects in the ground.

The essential requirements shall be specified by the Principal, often in consultation with the geophysical survey Contractor, and shall include the following:

- test method;
- required depth of penetration (either absolute depth or a strata boundary to be identified);
- tools and configuration;
- processing and reporting of results;

- submission by (geophysical) survey Contractor of detailed work plan, including quality aspects such as use of filters and software, personnel qualifications, expected accuracy of results for a range of site conditions, interpretative constraints.

The detailed work plan shall incorporate appropriate geophysical techniques, selected in the light of sufficient site knowledge.

The technical performance of geophysical methods is affected by:

- contrast in physical properties of the ground (high contrast gives a good reflection, low contrast, transitional changes etc. give a poor reflection and are hard to detect);
- ground water level (ground probing radar is often poor below GWL);
- penetration (deep penetration will give low resolution of surface deposits);
- equipment used : frequency, power, etc.;
- signal filters, interpretation software;
- sea conditions and water depth for boat mounted work;
- sea-bed surface deposits (gravel and debris can prevent signal penetration);
- vertical and horizontal resolution;
- signal-to-noise ratio.

8.3 INTERPRETATION AND REPORTING

Interpretative analysis should start at the time of data acquisition and should continue as an iterative process throughout the project. It shall be carried out by appropriately skilled personnel from the geophysical Survey Contractor. Interpretation of geophysical survey data involves some prior knowledge of the ground conditions for calibration purposes. Selected site investigation data shall therefore be copied to the (geophysical) Survey Contractor as early as possible.

Reporting requirements shall be specified by the Principal. These shall include:

- type and scope of report;
- number and scale of plans and cross sections to be produced;
- location of cross sections to be produced;
- details of (3D) computer model required, if any;
- details of required format for electronic data submission.

9. GEOTECHNICAL INVESTIGATIONS

Geotechnical investigations are also referred to as “ground investigation” or “soils investigation”.

Geotechnical investigations contain 4 basic stages:

- field survey;
- laboratory testing;
- interpretation;
- reporting.

Interpretation may be omitted from Contractor’s scope if only a factual report is required, however it has to be done by someone in order to transform raw data into design parameters and criteria.

9.1 OBJECTIVE

The objective of a geotechnical investigation is to provide information on ground and groundwater required for a project and to verify previously collected (desk study) data. The information gathered shall be sufficient to provide an adequate level of geotechnical design data for the relevant project stage(s).

The extent of the investigation should be sufficient to cover (or allow reasonable extrapolation over) the entire project area and should embrace both ground and groundwater. It should cover properties and behaviour under the full range of conditions that can be expected on the site. These include:

- temporary or permanent changes as a result of the construction project or natural phenomena such as: changes in stress and associated strain, changes in water content and associated volume changes, changes in groundwater level and flow pattern, changes in temperature and electrical potential, and changes in soil properties such as strength and compressibility;
- effects of earthquakes e.g. liquefaction;
- aggressivity to materials placed in the ground, such as concrete and steel;
- potential borrow areas and quarries for construction materials (fill and aggregates).

9.2 FIELD SURVEY

9.2.1 General

The common tools or techniques available for a geotechnical field survey are:

- boreholes and associated logging, sampling and testing (see 9.2.3);
- trial pits and trenches and associated logging, sampling and testing (see 9.2.4);
- penetration tests, such as the cone penetration test (CPT) and dynamic penetration tests (DPT) (see 9.2.5);
- piezometers (see 9.2.6).

Other tools which are fairly common are:

- Open drive samplers;
- hand augers;
- plate load tests;
- in-situ CBR tests;
- field vane tests (commonly called shear vane tests);
- (self-boring) pressuremeter tests.

There are many other less common techniques which may be of benefit for particular projects.

In planning a cost-effective investigation, relative costs of different elements shall be taken into account. The most expensive investigations are usually boreholes, followed by CPTs then trial pits and (lightweight) penetrometers. The amount of laboratory testing is a balance between spending money on field work or in the laboratory. Typically the amount of laboratory work will be highest per borehole in marine projects, where the unit price per borehole is also highest.

The cost of a CPT is generally much less than the cost of a borehole including associated sampling and laboratory testing. Where the ground is suitable the bulk of the investigations should comprise CPTs. However, a geotechnical survey shall not rely on the results of cone penetration tests only, as correlations between CPT data, borehole results and laboratory test data are necessary for safe and efficient design. In general, the ratio of CPTs to borehole should not exceed 10, but in most cases should be less than about 5. The higher ratios apply to situations that suit CPTs and where reliable correlations between CPT data and laboratory test data are available.

9.2.2 Scope of investigation

The exploratory holes (boreholes, CPTs etc.) shall be located in a manner that provides sufficient information for the relevant stage(s) of the project. They must make a general geological site assessment feasible. More detailed information should be obtained at positions of important structures and earthworks, at points of special engineering difficulty or importance and where ground conditions are complicated. In the absence of other criteria, a regular array of exploration points may be used in the initial design of the field investigation. Sufficiently close supervision and flexibility should be provided to allow adjustments to be made to the initial pattern as the survey proceeds, in order to ensure that the objectives are met. If the (preliminary) plant layout is known, points can be concentrated in relevant areas, or the overall grid made finer in areas where construction will occur (and coarser elsewhere). For various constructions and project phases, minimum and recommended numbers of exploratory holes apply. Minimum numbers apply where the ground conditions are consistent and known, and the investigations conform that these conditions occur at the location of the relevant structure. For variable, complex or unknown

ground conditions, an increased amount of investigation should be carried out. Existing investigations are included in the amount.

Table 9.1 Numbers of exploratory holes recommended for various project phases

Area	Scouting/Feasibility		Preliminary		Detailed	
	Minimum	Rec. **	Minimum	Rec. **	Minimum	Rec.
Plant areas	None *	500 m grid	100 m grid	50 m grid	50 m grid	20-50m grid
Tanks	None *	500 m grid	tank centre	5 points ***	5 points ***	Table 9.2
Roads	None *	500 m grid	200 m	100-200 m	100-200 m	50-100m
Pipe tracks	None *	500 m grid	200 m	100-200 m	100-200 m	50-100m
Pipe racks	None *	500 m grid	100 m grid	50-100m	50-100m	25-50 m
Pipe bridges	None *	500 m grid	100 m grid	each end	each end	1 to 3 at each end
Buildings	None *	500 m grid	100 m grid	50 m grid	50 m grid	25-50 m

* Use published information

** Increase amount if project is fairly definite

*** centre plus rest evenly spaced round outside

Table 9.2 Recommended number of exploratory holes for tanks

Tank Diameter (m)	Number of exploratory holes	Location of exploratory holes
< 10	1	Centre
10 - 24	5	Centre, plus 4 evenly round perimeter
25 - 49	9	Centre, plus 8 evenly round perimeter
50 - 59	13	Centre, plus 12 evenly round perimeter
60 - 90	17	Centre, plus 4 evenly round circle of half diameter, plus 12 evenly round perimeter
> 90	23	Centre, plus 6 evenly round circle of half diameter, plus 16 evenly round perimeter

For tanks, not all holes have to go to great depth. Typically it is sufficient for just the central hole and four of the outer holes to go to the full depth at which stresses from the tank will influence tank performance (settlement, stability). Remaining holes shall penetrate to the base of key settlement sensitive layers.

The spacing of exploration points for structures depends on many factors, including the variability of the ground and the availability or absence of relevant geophysical data. A preliminary investigation usually involves widely spaced exploration points to establish the general geological conditions, the suitability of different methods of investigation and the general groundwater conditions. A relatively wide spacing is often appropriate where complementary geophysical survey results are available. A minimum of three exploration points applies to structures, unless other reliable information is available in the immediate vicinity.

In some cases, exploratory holes should be offset so as not to interfere with subsequent construction, either because the ground disturbance will have an adverse affect, or because monitoring will be carried out at the point (e.g. by means of piezometer).

The need for and the number and spacing of piezometers will depend strongly on the topographical and geological nature of the site, the complexity of the groundwater regime and the ground water quality monitoring requirements.

For foundations of structures, the depth of exploration should be at least one and a half times the width of the loaded area, unless the imposed stress change becomes insignificant when compared with the strength and stiffness of the ground at a lesser depth, e.g. in rock. The required depth of exploration should be considered relative to the base of the foundation or raft. For foundations near the surface, the loaded area is considered as:

- 1) the area of an individual foundation, or
- 2) the plan area enclosed by two or more structures where the spacing of foundations is less than about three times their width or where the floor loading is significant, or
- 3) the area of a foundation raft.

Selection of exploration depths for pile foundations generally requires a preliminary analysis of possible pile types and lengths. An exploration depth equivalent to a minimum of 4 pile diameters below the lowest credible pile penetration is generally adequate for single piles. Pile groups will stress the ground below the pile tip to a greater depth than an individual isolated pile. The depth of exploration should be equivalent to about the width of the pile group below the expected lowest pile penetration. Special consideration should be given to made-ground, weak compressible soil and rock. Made-ground and weak compressible soils may not contribute to pile skin friction and may add downdrag to the pile. Checks on the continuity and weathering of rock are important for piles end bearing in rock. If voids may be present below pile toe level, deeper investigations may be called for.

The depth of exploration for earthworks, slopes and embankments should be sufficient for (slope) stability analysis (e.g. shear failure through weak ground strata), to assess ground movements (settlement, heave, horizontal displacement) and to establish groundwater conditions, including possible seepage or artesian water.

9.2.3 Boreholes

Boreholes are feasible in practically all types of ground and to any depth, both on land and over water. Borehole activities comprise borehole drilling, logging, in-situ testing sampling, installation of monitoring equipment, monitoring and backfilling. The quality of samples that can be obtained from boreholes is summarised in Appendix 4.

The common drilling techniques projects are:

- Open-hole drilling: a drilling method whereby all material within the diameter of the borehole is cut, such as open-hole rotary drilling, cable percussion drilling and auger drilling. No samples are obtained, it is simply a means to progress the hole to a depth required e.g. for testing or sampling.
- Cable percussion drilling: a drilling method whereby ground at the bottom of the borehole is broken up by percussive action of a bailer, clay cutter or chisel, and brought to the surface by the bailer or clay cutter. During drilling heavily disturbed samples can be obtained in clays. In non-cohesive ground, fines are lost and samples are not representative. Once depth is obtained samples can be taken by conventional means (e.g. open drive U100 sampler, piston sampler) or one of the wide range of in-situ tests available for boreholes tests done (e.g. SPT, pressuremeter, shear vane). Cable percussion drilling is typically done from a tripod rig.
- Auger drilling: an open-hole drilling method whereby ground at the bottom of the borehole is cut and brought to the surface by auger flights. Disturbed samples are obtained during drilling. Hand augers are often used for preliminary and small scale investigations, including for the gathering of samples for (preliminary) ground contamination assessment.
- Core drilling: a rotary drilling method that cuts out cylindrical ground samples. In soils, generally used as an open-hole drilling technique, although good quality cores of (stiff) clays are possible. Core drilling is the normal drilling method in rock. Cores are

obtained, the quality of which depends on the equipment and flush used, the driller's skill and the nature of the rock. In general good results (high core recovery, few drilling induced fractures in the core) should be obtained, but poor core recovery may be inevitable in:

- broken rock and gravel (lumps may be pushed round under the bit resulting in little recovery and slow progress);
- the first one or two metres of rock (as these are generally weathered and heavily fractured);
- cemented sands and very weak sandstone (tend to break down to sand even with very good drilling);
- very soft rocks containing hard lumps (e.g. chalk with flints - hard particles rotate in matrix rather than being drilled through).

Improved results can often be obtained by:

- changing to triple-tube core drilling;
 - shorter drill runs;
 - care extracting core from core barrel;
 - use sharper or diamond bit;
 - change speed of drilling, down-force, etc.;
 - other drill flush fluid.
- Water jetting: is also used as an open-hole drilling method, particularly in sands, soft silts and clays.
 - Hollow stem auger drilling: particularly used to obtain (disturbed) samples for environmental purposes.

The minimum requirements shall be specified by the Principal and shall include the following:

- minimum acceptable core recovery (total core recovery, solid core recovery and rock quality designator) and steps to be taken if it is not obtained and optionally drilling technique;
- required depth or termination criteria;
- minimum internal diameter of borehole, usually implicit because of minimum requirements for in-situ testing and sampling;
- programme of sampling and in-situ testing;
- in-situ test technique(s) and sampling technique(s);
- minimum internal diameter of sampler and cores, typically a minimum of 50 mm for tube samples but preferably at least 60 mm for rock coring samples and 70 mm for soil samples (larger for weak rocks and easily disturbed soils);
- minimum and maximum lengths of sample tubes, typically 300 mm minimum and 900 mm maximum for conventional tube samplers and typically 0.5 m minimum and 3 m maximum for core barrels;
- method of borehole backfilling, normally cement-bentonite mix;
- site laboratory activities including on-site logging and testing, sample/core photography;
- sample handling, storage and transport (temperature, shock and vibration), maximum time between sampling and laboratory testing, place and duration of retention of samples not used for testing.

In order to meet these requirements, the site investigation Contractor shall give details of the following with his bid:

- drilling technique, expected depth and termination criteria, usually agreed in terms of target depth and drilling equipment capabilities (torque, thrust etc.);

- methods for sample handling, transport, storage, etc.;
- how holes will be kept fit for purpose (e.g. stability, suitability for piezometers).

Recommended normal sampling programme:

- Cohesive soils: alternate “disturbed” and “undisturbed” samples at 1 m depth intervals. If SPT testing is regularly used in the ground sort, replace the “disturbed” samples with SPT tests.
- Non-cohesive soils: SPT tests at 1 m intervals. Replace some SPTs with bulk disturbed samples if drilling method allows.
- Rock: rotary coring with double-tube core barrel, to be provided with internal split liner (also called triple-tube core barrel) if total core recovery is less than 90%. If recovery is low because rock is broken or weak, carry out RPT (rock penetration test, similar to SPT but with solid cone) at end of each core run.

Note on SPT testing:

BS 5930 suggests a maximum of 50 blows before stopping. This should be changed to the following criteria:

- 50 blows during seating drive of 150 mm, or
- 50 blows for any 150 mm penetration increment, or
- 100 blows total.

The depth at which the results are reported shall be taken into account when interpreting the results.

NOTE: SPT results are often reported at the depth of the start of the seating drive rather than at the centre of the test section.

Backfilling of boreholes shall be carried out in a manner which seals the boreholes and prevents them from being a potential path for contaminant migration; and shall avoid leaving either a “hard” spot or a “soft” spot at the location. When clay layers are sealed between aquifers, the borehole shall be filled and sealed with clay-bentonite pellets or an appropriate powder/soil mixture. The procedure shall be subject to approval by the Principal.

9.2.4 Trial pits

Trial pits are generally used for shallow (up to 3 m depth) exploration on land. The depth limit is defined by the reach of the excavating machine. When a tracked machine is used, pits to 5 m depth are common. Deeper than 5 m is exceptional. Trial trenches are a special category of trial pits, being much longer and intended to show the ground structure in an area. Trenches are commonly dug with backacters, bulldozers or other excavating machines. Small pits (e.g. for exposing utilities or foundations) are often dug manually.

Trial pit activities for a geotechnical survey comprise trial pit excavation, logging, photography, (optional) in-situ testing and sampling. An important benefit of a trial pit is the exposure of a relatively large volume of ground and, in particular, ground structure. The large area makes identification of potential contamination and made ground simpler. Trial pits often provide a quick and cheap method of initial site appraisal (possibly combined with a few boreholes).

The procedure for a trial pit includes the following steps:

- assignment of trial pit details such as location, target depth, sequence of sampling;
- excavation and logging, such as stratigraphy, groundwater inflow, trial pit size and depth, excavation observations;
- shoring-up if required for entry;
- in-situ testing, including test depth and test parameters;
- sampling and sample handling;

- backfilling and compaction of the trial pit;
- data processing.

A range of in-situ tests are available for use in conjunction with trial pits. Examples are hand-held dynamic probing, hand-held in-situ vane testing and the in-situ density test by sand replacement. The more common sampling techniques are disturbed sampling and block sampling. Drive sampling and/or push sampling of an open-tube sampler may be feasible using manual equipment. Trial pits can also be used for tests for efficiency of soakaways for rain water drainage, but this is only relevant to areas which will never receive contaminated water.

For safety reasons, personnel should not descend into trial pits that are deeper than about 1.2 m. Logging, sampling and testing below this depth shall be from the ground surface by visual observations of the sides of the trial pit and by inspection/sampling of the excavated spoil. If the project requirements call for detailed logging, sampling and testing of deep trial pits, then temporary support by shoring or other means is necessary. Before deeper pits are entered, the possible presence of gas or fumes shall be considered. If there is any possibility of this (e.g. if there is hydrocarbon contamination of the ground) pits should not be entered without adequate precautions (e.g. safety man, breathing gear).

The essential requirements to be specified by the Principal shall include the following:

- trial pit depth, defined in terms of target depth, excavation equipment and termination criteria (e.g. instability, excavation refusal on rock);
- programme of sampling and in-situ testing (depth interval typically 1 m for samples, less for hand shear vane (HSV). If existing shear planes are being investigated, HSV at 0.1 m vertical intervals). Pilcon-type hand shear vane is preferred;
- in-situ test technique(s) and sampling technique(s);
- photographs of representative excavated face(s) of the trial pit;
- method of trial pit backfilling and compaction (badly compacted backfill is a hazard for foundations, roads and potentially for groundwater);
- site laboratory activities including on-site testing, sample/core photography;
- sample handling, storage and transport (shock and vibration), maximum time between sampling and laboratory testing, place and duration of retention of samples not used for testing.

9.2.5 Cone Penetration Test (CPT)

Cone penetration tests are suitable for most soil types. However penetration is typically poor in boulder clays, fill containing large debris, gravelly and rocky soils, and very dense sands. Penetration in soft rock is sometimes feasible, but to limited depth. CPT systems for land and over-water use are available.

A cone penetration test (CPT) involves in-situ measurement of the grounds resistance to continuous penetration at a steady slow penetration of a series of push rods having a cone at the base. CPT apparatus and procedures are standardised for both friction-cone and piezo-cone penetrometers equipped with internal sensors for continuous measurements at a penetration rate of approximately 20 mm/s. Other terms used for a CPT are "static cone test" and "Dutch cone test".

Modern equipment includes electric cone penetrometers, however in some parts of the world mechanical cone penetrometers are still in use. These should only be used with extreme caution and where there is clear justification (e.g. unacceptably long mobilisation time to get an electric cone penetrometer). They are not normally acceptable for main or detailed investigations for large projects. If part of the site data comes from mechanical CPTs, correlation with electric CPT testing close by is recommended for a number of holes.

The preferred equipment is thus the electric friction cone penetrometer with an inclinometer in the cone penetrometer. The accuracy should be: cone resistance: 0.5 MPa or 5% (whichever is worse); sleeve friction: 0.05 MPa or 20%; cone penetrometer inclination: 5°;

CPT depth (vertical, corrected): 0.2 m or 2%. 200 kN (20t) reaction force is preferred, as lighter (100 kN) units are often unable to reach adequate penetration.

The essential requirements shall be specified and shall include the following:

- Required depth and/or termination criteria. Common termination criteria for CPT systems are (whichever is earlier):
 - target depth;
 - surface-based thrust of 200 kN (or 100 kN for shallow penetration) (equipment limit);
 - cone resistance of 80 MPa (to avoid damage);
 - sleeve friction of 0.5 MPa (to avoid damage);
 - cone penetrometer inclination of 20° or 3° sudden increase in combination with a high cone resistance. (to avoid jamming or damage).
- Application of a friction reducer and/or push rod casing and drilling equipment capabilities (torque, thrust etc.).
- Ordinary or special (e.g. piezo-cone) penetrometer.
- Special testing, such as pore pressure dissipation, electrical conductivity, shear wave velocity.
- Optional drill-out to the CPT termination depth and extension of testing below the initial termination depth with the use of a guide casing (this is standard procedure for down-hole or mounted equipment which has a typical stroke length of 3 m).
- Method of CPT hole backfilling (not normally required).
- Any special processing or reporting of results (net cone resistance, pore pressure ratio, computer-based interpretation; digital data format).

CPT hole backfilling is uncommon, but cement-bentonite backfilling shall be considered where an open hole presents a threat of the spreading of environmental pollution.

9.2.6 Dynamic probing

Dynamic Probing (DP) involves in-situ measurement of the ground's resistance to continuous dynamic penetration of one or more push rods having a cone at the base. A hammer of a given mass and falling height is used to drive the cone. Typical weights and drops range from 10 kg dropping 50 cm up to SPT weight and drop. Well known types are the DIN, Borros, Mackintosh. All these types are portable and can be manhandled through ordinary doorways. The light dynamic probe (DPL) represents the lower end of the mass range (10 kg) of dynamic penetrometers. DPL apparatus and procedures are standardised. DPL is generally suitable for most soil types. Penetration varies with type and impact energy, and varies from a few metres up to 12 m in clays and chalk. Significant penetration into hard soils is unlikely. DP is cheap and has a high production rate (about 60 linear metres or more per day). Uses are:

- probing soft soil thickness;
- rapid correlations of relative density/strength of soils (done in combination with e.g. a few boreholes and SPT testing);
- initial investigations in areas where no access is available to drilling rigs or CPT trucks.

The requirements shall be specified by the Principal and shall include the following:

- depth and termination criteria, usually agreed in terms of target depth, maximum blow count per 100 mm penetration, total number of blows and rebound of falling weight;
- application of any friction reducing measures;
- torque measurements for rod rotation;
- special processing of results (corrections for friction, unit point resistance, dynamic point resistance, correlations to SPT tests).

Common termination criteria for DPL testing are (whichever is earlier):

- target depth;
- 50 blows per 100 mm;
- rebound per blow of more than 50% of penetration per blow;
- ringing sound combined with no significant penetration;
- total of 250 blows.

Special types of dynamic penetrometer are:

- open drive samplers: continuous samples are taken with the aid of an open-sided tube. Drive is typically by Kango hammer or similar;
- small proprietary brand piezometers which can be installed with the aid of hollow rods and a special tip.

9.2.7 Piezometers and monitoring wells

Piezometer measurements consist of determining the groundwater level or the pore pressure in the ground. Some piezometers can also be used for in-situ permeability measurements. There are two main groups of piezometers: open and closed systems.

Characteristics of open measurement systems are direct contact between groundwater and atmosphere: groundwater flows into or out of the piezometer as the groundwater level varies. Piezometers are constructed made by installing open pipes or pipes provided with filters and (if required) transducers and data-loggers to record the pressure at the level of the tip. Measurements of water level are typically made in a borehole, pipe or hose using a dipmeter. The response time is usually adequate for high-permeability soil but may be inadequate for low-permeability soil. Most common are standpipe piezometers.

Characteristics of closed measurement systems are (1) prevention of contact between groundwater and atmosphere/fluid within the piezometer, and (2) pressures are recorded by a transducer. They are therefore readily suited to readout units being some distance from the installation. As there is no water flow in and out, closed measurement systems allow a short response time in most soil types. Examples include hydraulic, electrical and pneumatic piezometers.

Piezometers/monitoring wells are increasingly used for environmental monitoring and for groundwater sampling. If water samples may be required, a relatively large diameter open piezometer is needed to facilitate sample abstraction. This typically will give a slow response time. If dual purpose monitoring wells are intended, it must be checked that the response time is sufficient for groundwater level monitoring.

The response time of a groundwater piezometer is an important consideration for selection of a particular type of piezometer. It depends on the quantity of water required to enter the system (including pipes and/or tubes) and on the permeability of the ground.

Typical standpipe piezometer/monitoring well activities are as follows:

- drill borehole;
- grout up to below response zone;
- install bentonite plug up to base of response zone;
- install standpipe piezometer/monitoring well and sand/gravel filter in response zone;
- install bentonite plug above response zone;
- backfill rest of borehole;
- install lockable cap / surface protection and marker;
- monitor water levels;
- decommission / grout up.

The requirements shall be specified by the Principal and shall include the following:

- type of piezometer/monitoring well, depth of response zone(s) (or strata to be monitored);

- functional requirements (e.g. how long it has to work, response time, special requirements such as use for permeability testing);
- general geometry of borehole and piezometer materials, sometimes implicit because of minimum requirements for monitoring and sampling, including special requirements if groundwater contamination may be aggressive to normal materials;
- permissible drilling water additives;
- programme of monitoring and sampling including reporting requirements;
- physical protection (cover, fence, marker, protection against freezing);
- associated water level measurements required in nearby surface waters (lake, sea, river) to see if groundwater mirrors levels there;
- method of decommissioning (if required).

Unless there are other specific requirements, the following monitoring programme is recommended for piezometers/monitoring wells: for the remaining duration of a site investigation, monitoring is to consist of daily water level measurements until steady-state conditions have been reached, followed by weekly monitoring. After completion of the site investigation, weekly until performance is clear (e.g. static or changing water level, amount and speed of response to rainfall, etc.), then monthly. Long-term monitoring usually requires clear arrangements, including responsibility and access. All monitoring programmes should allow for redundancy because of practically inevitable loss of piezometers due to incidents. The amount of redundancy varies with importance of the measurements and activities in the area. A judicious choice of piezometer/monitoring well location is particularly important for reducing the loss rate.

Interpretation of measured water levels requires some caution. Unless data-logging is used, measurements are made at a small number of discrete points in time. Misleading results can come from:

- the standpipe piezometers/monitoring well is in low-permeability soil reacting too slowly;
- rapid changes in water levels in permeable soils in inclement weather;
- piezometers which intercept several different water bearing strata with different piezometric heads;
- seasonal and other environmental influences, including construction activities;
- tidal effects;
- river flooding effects;
- variations with rainfall;
- blockages;
- surface or rainwater leaking into piezometer pipe directly (faster than it can seep into ground via piezometer tip) e.g. via leaky cover;
- barometric pressure.

It is clear from the above that in order to be able to interpret the results, piezometer/monitoring well measurements should be accompanied by records of weather, tide levels, other influences, etc.

9.2.8 In-situ testing

In-situ testing often gives more reliable results than laboratory testing (because the scale is larger, and the correct stress field applies) or may be cheaper (no need to obtain samples and transport them to the laboratory). Commonly applied tests are:

- CPT, SPT, penetrometer, shear vane, etc.;
- pressuremeters: useful for strains, particularly lateral strains, in-situ ground stresses, deformation predictions. Extensively used in France, less elsewhere. Common for establishing in-situ stresses in overconsolidated soils;

- plate bearing tests: measures load-settlement parameters for plates up to approx. 1000 mm diameter;
- trial embankments (pre-load can be considered to be a trial embankment): give settlement/consolidation data as well as improving the ground). Particularly useful where scale effects make smaller tests such as laboratory tests and plate bearing tests of limited value;
- resistivity testing (provides data for design of earthing systems);
- anchor tests;
- CBR tests (for road construction);
- permeability tests (in boreholes) are much better than laboratory tests, as the larger scale allows for structure effects;
- pumping tests from wells;
- grouting trials e.g. grout take of different grouts, change in ground conditions after grouting (voids, permeability);
- rock blasting trials to see amounts of different sizes that can be obtained, etc.;
- density, construction control - e.g. full scale trials of compaction methods;
- etc.

9.3 LABORATORY TESTING

9.3.1 General

The purpose of laboratory testing is to provide data, complementary to the field investigations, to define the (ground) conditions sufficiently to allow design to be accurate enough for the requisite project stage(s). In general laboratory work is concerned with index tests, which are used to classify areas or layers of soil into groups, and tests to describe the physical properties of the soil (strength, compressibility, permeability, etc.) at the various stress ranges that may occur during the lifetime of the project.

The common groups of test methods available for a laboratory test programme are:

- classification: for identification and description of ground (also referred to as index testing);
- strength tests on soil and rock;
- compressibility and consolidation tests on soil;
- compaction tests on soil: for construction control;
- permeability tests on soil;
- tests on aggregates for road pavement and concrete (strength, abrasion resistance);
- mineralogy / reactivity tests for aggregates for concrete;
- tests on rock for erosion control (e.g. for durability under wave action);
- chemical: ground and groundwater - for aggressivity to foundations and for contamination identification/monitoring.

Test requirements are generally set out in BS 1377. ASTM Volumes 04.02, 04.08 and 04.09 include standards for a wider range of techniques, including standards for less common laboratory test methods. These standards should be used where no BS standards are available.

9.3.2 Laboratory test programme

The laboratory test programme shall take account of the geotechnical aspects of the project, including ground types, stratigraphy and changes imposed by the construction project itself or by natural processes.

The minimum amount of testing shall be sufficient to establish the requisite parameters to an acceptable degree of accuracy. A balance should be struck between the cost of the testing and the benefits of better parameter definition (lower partial factors, less uncertainty, saving in construction costs). The amount of testing will also reflect the amount that is already known about a given soil type. If there is extensive literature on a soil type and its properties, it may be sufficient simply to confirm that the soil corresponds to the published data, and with the help of index properties other design parameters can be derived from literature values. This is particularly useful for small work items with tight programmes.

As a general guideline 3 or more tests should be carried out for any given parameter in a particular soil type. Fewer than these are not representative

Ground samples shall be inspected visually to establish a preliminary stratigraphic profile. Visual inspection should be supported by simple manual tests to classify and identify the ground. The strata relevant for the geotechnical aspects of the project shall be selected to enable specification of the type and number of tests for each ground unit. The number of tests depends on factors such as the number and quality of available samples and the availability of comparable experience. The tests should normally be run on representative specimens. The use of any non-representative specimens shall be reported together with the test results.

The feasibility of a particular laboratory test, in terms of the agreed sampling practice and sample handling, depends on factors such as ground type, available amount of sample material and sample quality. Usually, a reasonable estimate of test feasibility is possible at the time of sampling. A further refinement is possible in the laboratory prior to testing and,

in some cases, only after testing. The limitations of feasibility estimates may lead to rejection of samples for testing upon inspection in the laboratory or may result in adjustments to the testing programme. Appropriate comments on the test programme and test results shall be made.

BS 5930 recognises 5 classes on the basis of feasibility of specific geotechnical identification and laboratory tests on ground samples. Appendix 4 includes a summary of these classes. The effects of sample disturbance shall be taken into account when planning the test programme and when interpreting the results. Appendix 4 summarises the feasibility of various sampling techniques for achieving a particular sample class.

Consideration shall be given to more advanced tests or adjustments to standard practice, in case of unusual ground conditions or complex aspects of geotechnical design.

9.3.3 Classification, identification and description of ground

Classification, identification and description of ground shall be according to BS 5930. More specific description systems may be appropriate for unconventional soil or rock, such as peat and cemented carbonate sediments. The test programme shall cover all available samples.

Laboratory classification tests shall be conducted to assist in classification, identification and description of ground. Also, the results of classification tests can help to explain the behaviour of ground as observed in specific in-situ and laboratory tests. The common laboratory classification tests are (comments between brackets indicate typical quantities of tests per stratum):

- water content (most samples or every 2 m);
- bulk density (most samples or every 2 m);
- particle density for both soil and rock (1 or every 10 m);
- particle size distribution (4 or every 5 m);
- Atterberg limits of cohesive soil (3 or every 5 m);
- carbonate content (2 or every 10 m);
- microscopic inspection of rock (2 or every 10 m);

Some projects may require classification tests for soil dispersion and frost susceptibility.

9.3.4 Chemical tests on ground and groundwater

Chemical tests shall be conducted to provide a basis for assessment of potential detrimental effects of ground and groundwater on concrete, steel and the ground itself. This section does not cover chemical testing for environmental assessment or monitoring purposes.

The common chemical tests for soil and rock are organic content, carbonate content, sulphate content, chloride content and pH value (acidity or alkalinity). The common chemical tests for groundwater are sulphate content, chloride content, total soluble salts, total dissolved solids and acidity (pH value). Test quantities depend strongly on the project requirements and the nature of the ground. A set of 3 measurements for a critical stratum may be used as an initial guide.

In acid soils other testing such as sulphide content will be necessary to establish the extent of the problem.

9.3.5 Compressibility tests on soil

Compressibility tests shall be conducted where information about deformation, compression, consolidation or swelling characteristics of soil is required (e.g. where settlement may be an issue). The standard test type is the incremental oedometer for cohesive soils.

Results of compressibility tests have to be used with care. In particular:

- Actual soils, particularly sediments, are usually layered and have much higher horizontal than vertical permeability (factor 5 to 100, typically about 10). The oedometer test normally measures vertical permeability of samples 20 mm thick, so generally underestimates permeability and settlement times by a large factor. In practice this effect is normally favourable, but can be wasteful - e.g. preloads planned in for 6 months when 1 to 2 weeks might be enough.
- Only intact homogeneous specimens are normally tested; if the ground is very variable, these may be unrepresentative.
- In some soils, particularly peat and very organic soils, permeability and speed of consolidation are very sensitive to disturbance, stress and stress history (e.g. vegetation rots to leave open passages which crush when loaded and do not re-open).

The essential requirements shall be specified and for an incremental oedometer test on undisturbed cohesive soil shall include the following:

- type of test;
- supplementary classification tests;
- specimen orientation and drainage type;
- vertical stress increment for saturation for specimens with collapse or swell potential;
- stress increments and decrements;
- stress increments for time-settlement analysis.

9.3.6 Strength tests on soil

Strength tests shall be conducted where further information about drained (effective stress) and undrained shear strength parameters are required.

Common strength index tests for undrained shear strength of cohesive soil are:

- laboratory vane;
- torvane and pocket penetrometer;
- unconfined compression;
- unconsolidated undrained compression;
- consolidated undrained compression (normally with pore pressure measurement, giving drained and undrained information).

The unconsolidated undrained compression test gives more reliable results than the first three strength index tests, although it may not represent field conditions because of stress relief and sample size effects. A typical test programme includes one or more of the strength index tests on each of the available suitable samples.

The essential requirements to be specified for strength index tests shall include the following:

- type of test;
- supplementary classification tests;
- specimen orientation;
- sample extrusion prior to testing or testing in sample tube for laboratory vane;
- cell pressure for unconsolidated undrained compression.

Effective stress testing is normally carried out on undisturbed samples of cohesive soils in a triaxial test apparatus, sometimes in a shear box. Common effective stress strength tests are consolidated undrained with pore pressure measurement (CUPP) and consolidated drained (CD) triaxial compression and drained direct shear box. (Pore pressure measurement is a standard feature of effective stress testing, and CUPP tests are sometimes referred to as CU tests). The common consolidation practice for triaxial compression is isotropic consolidation.

It is important that sufficient testing be carried out to establish parameters over the range of stresses that will occur on site over the lifetime of the project.

As non-cohesive soils cannot normally be recovered in an undisturbed state, remoulded samples may be used. It is recommended that samples be tested loose (to give lower bound strength), recompacted densely (to give upper bound strength), and if feasible compacted to roughly the same state as in the field (to give "best guess" strength). Remoulding in a shear box is often used because it is easier than making test specimens for triaxial tests.

The essential requirements to be specified for an isotropically consolidated undrained triaxial compression test on undisturbed cohesive soil shall include the following:

- type of test;
- supplementary classification tests;
- specimen orientation;
- consolidation pressure;
- cell pressures for testing.

The essential requirements shall be specified and for an isotropically consolidated drained triaxial compression test or a direct shear box test on re-compacted cohesionless soil shall include the following:

- type of test;
- density index test;
- supplementary classification tests;
- practice of specimen preparation;
- test practice and consolidation pressure: single specimen and single consolidation pressure or three specimens and three consolidation pressures.

The cohesion intercept (c') in a triaxial test is heavily influenced by the test procedure, particularly the degree of saturation of the specimen, type and degree of saturation of end plates and speed of testing. Values of c' significantly above zero should only be used if supported by other (e.g. published) data.

If soils will be subject to cyclic loading, particularly if there are complete stress reversals, degradation and reduction in strength and stiffness may occur. This is particularly pronounced in overconsolidated clays subject to cyclic lateral loads. Cyclic load testing (normally triaxial testing) can be carried out to investigate the effects. It is, however expensive and time consuming, and only a few laboratories have the facilities, so it is normally restricted to locations where the effects may be critical or where improved data accuracy will permit savings compared with adopting a conservative design approach based on conventional testing and published results.

9.3.7 Compaction tests on soil

Compaction tests shall be conducted where further information about the relationship between dry density and water content is required for a given compaction effort. Compaction tests can provide a basis for specifying field compaction requirements. Common compaction tests are the Standard (Proctor) and the Modified Compaction tests.

California Bearing Ratio (CBR) tests may be conducted where further information about the potential strength of subgrade, sub-base and base course materials is required for specific empirical design methods for roads, railways and airfield pavements. There are also special tests for specific ground types which do not behave in the same way as most soils, e.g. the MCV test for chalk fill.

9.3.8 Permeability tests on soil

Permeability tests shall be conducted where further information about the coefficient of permeability is required. The soil structure is very important in determining actual permeability of the ground. Horizontal permeability is commonly much greater than vertical, so the test orientation is critical. For all materials large scale field tests are recommended in preference to laboratory tests. The simplest to perform is a falling head test in a borehole or piezometer.

The accuracy of permeability testing (in terms of how representative it is for the site) is rarely more accurate than a factor of 3, except for uniform sand deposits. It is common to see a range of an order of magnitude for tests in (samples from) the same stratum. In the evaluation of test results, a relatively large variation in coefficient of permeability should be expected, for example due to small changes in stress, void ratio, structure (for example horizontal versus vertical permeability) and particle size. Furthermore, the degree of saturation can influence the coefficient of permeability up to as much as three orders of magnitude for some types of soil.

Least consistent are laboratory tests on low permeability clays. Here the reported results from several labs/test types for the clay may vary by several orders of magnitude. In addition, the intact specimens tested are small and often do not reflect the macro-structure, where thin higher permeability routes may dominate behaviour. Tests on remoulded samples are more consistent but less representative as structure is destroyed in the remoulding.

Permeability of a soil is not the same for different fluids. Tests carried out for water cannot therefore be assumed to apply for other substances. For hydrocarbons, viscosity has probably the largest effect, but different soils respond to different chemicals in different ways. When pore fluid or redox conditions change, changes of permeability with time will often occur.

Common permeability test types are the falling-head test on an undisturbed specimen of cohesive soil and the constant-head test on a re-compacted specimen of cohesionless soil. Common apparatus for these test types are the permeameter and the triaxial cell.

The essential requirements to be specified for a falling-head test on an undisturbed specimen of cohesive soil shall include the following:

- type of test
- supplementary classification tests;
- specimen preparation and orientation;
- consolidation pressure for triaxial apparatus;
- degree of saturation for triaxial apparatus;
- temperature correction.

Additional requirements to be specified for a constant head test on a re-compacted specimen of cohesionless soil shall include the following:

- density index test;
- test method and consolidation pressure.

9.3.9 Strength tests on rock

Strength tests shall be conducted where information about rock strength and deformation is required. Standard strength tests are for uni-axial compressive strength of intact rock material and the point load strength index. For strength of discontinuities, the direct shear box test on a rock discontinuity is often used. Rock strength test results vary greatly, and sufficient tests shall be carried out to allow meaningful statistical analysis of results. If this is not done a very conservative lower bound strength has to be adopted.

Some common limitations for strength tests on rock are:

- testing of intact rock material rather than more representative broken rock;
- mass discontinuities untested;
- a bias towards testing the more competent sections of rock material, as the less competent sections may be affected by poor recovery or none at all.

The essential requirements to be specified shall include the following:

- applicable external standard;
- supplementary classification tests;

- specimen source and depth, type (core, block, irregular lump), specimen orientation and dimensions;
- optional mean value and anisotropy index calculation;
- requirements for correlations (correlation between point load tests and UCS test results shall be provided);
- optional P-wave velocity measurement;
- optional axial and radial deformation measurements.

The essential requirements to be specified for a direct shear box test on a rock discontinuity shall include the following:

- type of test;
- supplementary classification tests;
- specimen orientation and dimensions;
- test practice and normal stress.

9.3.10 Testing of aggregates for road pavement and concrete

Common laboratory tests are:

- aggregate type, shape and surface texture;
- petrographic examination in particular to look for deleterious components;
- flakiness index;
- 10% fines value;
- aggregate impact value;
- particle density;
- water absorption values;
- particle size distribution;
- acid soluble sulphate and chloride ion contents;
- drying shrinkage.

BS 882 tests should normally be used.

9.3.11 Testing of rock for erosion control

Common laboratory tests are:

- petrographic examination, supplemented by ethylene glycol soaking or methylene blue absorption for detection of smectite minerals;
- particle size distribution;
- wetting and drying durability;
- freezing and thawing durability;
- sodium or magnesium sulphate soundness;
- apparent particle density;
- water absorption values;
- Los Angeles abrasion;
- splitting tensile strength;
- insoluble residue for limestone or dolomite.

9.4 INTERPRETATIVE GEOTECHNICAL ANALYSIS AND REPORTING

9.4.1 General

The Geotechnical Interpretative report provides the basis for ongoing geotechnical and foundation design for the project. It is generally not provided as a “rely-upon” document to Contractor, however is in practice used as a rely-upon document by Principal. For investigations carried out by Contractor it is normally a rely-upon document. It is vital that the interpretation is correct and if there are any limitations or uncertainties in the interpretation, that these are clearly documented in the report.

In order to get a good interpretative report, deliverables must be well defined. It is recommended that for procurement manhour estimates be required from contractors: these can be checked against input later; and used for comparison of different bids.

The interpretation and report shall take account of all available data, not just the latest field and laboratory work. The Principal should ensure that the report author receives copies of, or is given access to, all available data.

The essential requirements for interpretative analysis shall be specified and shall include one or more of the following items:

1. ground and groundwater characterisation;
2. evaluation of derived geotechnical parameters;
3. selection of geotechnical options (e.g. foundation alternatives, ground treatment required/possible methods, etc.);
4. selection of geotechnical approach, such as use of prescriptive measures or application of characteristic geotechnical parameters in a calculation model;
5. soil liquefaction assessment;
6. preliminary recommendations for foundation (e.g. tables of pile size and length against capacity);
7. stability of existing and proposed slopes;
8. a clear statement of how representative values and design values for parameters have been derived, whether design foundation loads are ultimate or factored, safety factors used, etc.;
9. seismic hazard assessment;
10. recommendations for additional studies: when the reliability of the conclusions and recommendations may be inhibited by lack of sufficient tangible evidence or experience.

Interpretative analysis for a geotechnical survey should include integration of geological and geophysical survey data. A close and active co-operation between the individual specialists concerned is often necessary at every stage, from preparatory work to drafting of the final interpretative report.

If a number of individual investigations are carried out, with separate (interpretative) reporting, a summary report shall be produced at the detailed design stage which sets out the ground characterisation, parameters etc., and their derivation, which will be used in design. Each report should comment on how data from previous reports is used and on any conflicts with earlier reports.

9.4.2 Ground characterisation

General ground characterisation is necessary before selection of geotechnical parameters for the ground and before evaluation of the results of specific tests and observations. Such ground characterisation comprises a general assessment of the structure, character and basic constituents of the ground (soil and rock classification). It shall include:

- geological and hydro-geological setting;
- 3D description of the ground (by means of text, cross sections, mapping of boundaries derived from geophysical surveys, and where required computer model);
- characterisation of the ground into discrete layers, sub-layers and areas, to each of which are assigned specific parameters (these parameters may depend on factors such as depth within a discrete layer);
- typical parameters for soil classification including particle size distribution, water content, carbonate content, Atterberg limits, unit weight, relative density and undrained shear strength (or typical parameters for rock classification including mineralogy, water content, unit weight and uni-axial compressive strength);
- details of which soil layer(s) are water-bearing, and the piezometric heads in each layer. Levels of bodies of open water;
- aggressiveness of ground and groundwater;
- existing features that may be relevant (e.g. existing landslides, site history, difficult access).

Options for ground characterisation may include additional evaluation of:

- soil liquefaction susceptibility;
- stability / landslide risk;
- collapse settlement possibilities.

9.4.3 Soil liquefaction susceptibility

Liquefaction represents a decrease of soil strength and stiffness caused by an increase in pore water pressure in saturated soil. It typically results in loss of bearing capacity, slope instability, settlements, failure of retaining structures, etc. The most common cause is earthquakes, but it can also be triggered by cyclic loads, vibrations or rapid stress relief. Loose poorly graded (single-size) saturated fine sands and coarse silt are most susceptible, although denser sands, loose unsaturated sands and some sensitive cohesive materials (e.g. "quick" clays) can also liquefy under some conditions. The susceptibility can be related to the magnitude of accelerations required to cause liquefaction. One objective of the site investigation is to identify zones which may be subject to liquefaction and the associated accelerations required to trigger it. Interpretative reporting should cover measures that can be taken to remove the problem.

The common empirical procedures rely on in-situ SPT, CPT and shear wave velocity tests to assess insitu relative density. These are supplemented by particle size analysis (including fines fraction) and Atterberg limits tests in the laboratory to see if soils fit into categories considered susceptible to liquefaction

For a given design peak ground acceleration (cyclic liquefaction) or in-situ shear stress regime (gravitational liquefaction), the following general procedure is common to analyses of either type of liquefaction:

1. general evaluation of ground conditions (stratigraphy);
2. material characterisation (soil type, grading, in-situ density and strength);
3. interpretation and analysis of in-situ tests;
4. interpretation and analysis of dynamic (cyclic) and static (gravitational) laboratory tests on reconstituted samples, if performed;
5. assessment of likelihood and extent of liquefaction for various magnitudes of earthquake, or for design earthquakes if known.

9.4.4 Geotechnical parameters

In-situ test and laboratory test measurements and other relevant (e.g. published) data provide a basis for obtaining derived, characteristic and design (calculation) values of geotechnical parameters. Empirical correlations and theoretical interpretation models are available for determination of specific derived geotechnical parameters (when they are used it shall be ascertained that they are valid for the actual soil and location). Statistical methods and/or engineering judgement are used to evaluate design parameters for geotechnical analysis from actual test results.

In general parameters should be derived on the basis of Eurocode EC7 and associated partial design factors used. The method of parameter derivation shall be clearly reported. Some codes (e.g. the Dutch NEN 6740) provide details on how the statistical analysis may be carried out.

When statistic methods are not used, and when soil parameters are derived for use with a total factor of safety design approach, the associated calculation method and recommended factors of safety shall also be reported.

Parameters which shall normally be determined are:

Index properties: w (mc), w_l (LL); w_p (PL); I_p (PI); PSD; e ;

Strength: ϕ' ; c' ; c_u ; (s_u); σ_c (UCS); point load strength

Compressibility: E ; C_v ; C_c ; C_r ; m_v

Stress state: OCR; p_c'

Density: D_R ; γ ; γ_d

Permeability: k ; k_h ; k_v

9.4.5 Geotechnical analysis

Geotechnical analysis in site investigation reports is normally only preliminary. DEP 34.11.00.12-Gen. shall be referred to for calculations which will be used for detailed design.

Typical subjects for geotechnical analysis in an interpretative report are as follows:

- 1) site preparation, earthworks and general drainage:
 - assessment of groundwater table fluctuation
 - aggressiveness of groundwater and soils;
 - site settlement due to placement of fill;
 - liquefaction sensitivity and possible compaction requirements;
 - roads and railways;
 - buried pipelines;
 - settlement behaviour;
 - influence of works/constructions on adjacent structures.
- 2) shallow foundations:
 - bearing capacities;
 - settlements (total and differential) to be anticipated;
 - time/settlement behaviour.
- 3) tank foundations:
 - total and differential settlements to be expected at tank centre and at tank shell;

- time/settlement behaviour.
- 4) pile foundations:
 - type of piles;
 - bearing capacity of piles (axial and lateral);
 - total and differential settlements expected for single piles, for pile groups;
 - driveability;
 - negative skin friction.
- 5) advice on foundations for vibrating machinery.
- 6) dolphins and jetties: (API RP 2A methodology may be used):
 - type of piles;
 - bearing capacity of piles (axial and lateral);
 - total and differential settlements expected for rows of piles and for a group of piles;
 - horizontal load-displacement behaviour of piles;
 - driveability;
 - long term behaviour/creep (vertically and horizontally).
- 7) slope stability
- 8) phasing of fill and preload (on soft soils)
- 9) ground improvements (e.g. to remove liquefaction potential or reduce settlements to acceptable levels)
- 10) limits of validity of design assumptions and of data interpretation.

10. HYDRO-GEOLOGICAL AND METEOROLOGICAL SURVEY

10.1 OBJECTIVE

The objective of a hydro-geological and meteorological survey is to verify and expand previously collected (desk study) information for assessment of the impact of hydro-geological and meteorological conditions on the project and the impact of the project on the existing hydro-geological conditions. Such impact assessment typically includes ground water control, water supply and flood risk. Results will normally be presented in terms of mean annual data and events with relevant return periods (e.g. 10-25 years for temporary works design; 100, 1000 and 10 000 years for detailed design).

The survey shall provide information including seasonal variation on:

- precipitation and snow cover;
- evaporation;
- temperature (air and ground) and ice formation;
- humidity;
- wind, including storm events;
- groundwater levels;
- aquifers and other water bearing layers;
- quality and variations in quality of groundwater;
- surface water run-off;
- open wells, where relevant;
- erosion due to surface run-off;
- existing drainage systems (or absence thereof).

A hydro-geological survey will overlap with a geotechnical survey, in particular with respect to groundwater.

10.2 FIELD SURVEY AND LABORATORY TESTING

Various field and laboratory survey methods are available for hydro-geological and meteorological data acquisition, where inadequate information is available from the desk study. A check shall be made that the minimum length of time required for the monitoring surveys is compatible with the project programme.

10.3 INTERPRETATIVE ANALYSIS

The requirements for interpretative analysis shall be specified according to the objectives of the survey.

The use of historic or data base information requires particular care for explanation of apparent anomalies such as missing data (for example no event may be recorded if a monitoring station is destroyed by an extreme event).

11. OCEANOGRAPHIC / HYDRAULIC SURVEY (RIVER, CANAL, LAKE OR SEA)

11.1 OBJECTIVE

The objective of a oceanographic/hydraulic survey is to obtain or verify and expand previously collected (desk study) information for assessment of the impact of the oceanic and hydraulic conditions on the construction project and the impact of the construction project on the existing hydraulic regime.

The survey shall typically provide information on:

- 1) For sea:
 - tidal ranges including highest astronomical tide (HAT), mean sea level datum (MSL) and lowest astronomical tide (LAT);
 - tides with reference to standard and secondary ports;
 - currents (prevailing current velocity and direction, tidal currents, etc.);
 - wave patterns including height, periods and duration;
 - general morphological data;
 - quantity and pattern of siltation/erosion;
 - ice conditions;
 - temperature;
 - salinity;
 - sediment concentration.
- 2) For rivers and canals as for sea, plus:
 - stream flow pattern and velocity;
 - expected variation in water level and discharge;
 - indications of siltation/erosion, bank and bed stability;
 - occurrence of sediment and floating debris.
- 3) For lakes as for sea, plus:
 - expected variation in water level;
 - occurrence of storms and high winds, and their effect on generating waves.

11.2 FIELD SURVEY

Various field and laboratory survey methods are available for hydraulic data acquisition, where inadequate information is available from the desk study. A check shall be made that the minimum length of time required for the monitoring surveys is compatible with the project programme - monitoring over a whole year or a complete season is a common requirement.

11.3 INTERPRETATIVE ANALYSIS

The essential requirements for interpretative analysis shall be specified by the Principal. These will normally comprise interpreting the data to give predictions for events with certain return periods, and for average conditions.

The use of historic or data base information requires particular care for explanation of apparent anomalies such as missing data (for example no event may be recorded if a monitoring station is destroyed by extreme event).

Particular consideration shall be given to potential hazards, such as:

- flooding, erosion and stability of the coast and river banks;
- impact of a tsunami, if applicable.

12. ENVIRONMENTAL SITE ASSESSMENT

Environmental site assessment is not covered by this DEP. The reader is referred to SIOP document OP 96-3026 'Recommended Practice for Environmental Site Assessment'. This recommended practice also applies where an environmental survey is combined with a geotechnical survey. Preliminary evaluation based on visual observation and information obtained does, however, form part of a desk study.

13. REFERENCES

In this DEP reference is made to the following publications:

NOTE: Unless specifically designated by date, the latest edition of each publication shall be used, together with any amendments/supplements/revisions thereto.

AMERICAN STANDARDS

Concrete and aggregates	ASTM Vol. 04.02
Soil and Rock (I) D 420 - D 4914	ASTM Vol. 04.08
Soil and Rock (II) D 4943; Geosynthetics.	ASTM Vol. 04.09

Issued by:
American Society for Testing and Materials
1916 Race Street,
Philadelphia Pa 19103
USA

BRITISH STANDARDS

Specification for aggregates from natural sources for concrete	BS 882
Methods of test for soils for civil engineering purposes	BS 1377
Code of practice for site investigations	BS 5930
Code of practice for earthworks	BS 6031
Code of practice for maritime structures, Part 1: General criteria	BS 6349-1
Code of practice for maritime structures, Part 2: Design of quay walls, jetties and dolphins	BS 6349-2
Maritime structures, Part 5: Code of practice for dredging and land reclamation	BS 6349-5
Part 7: Guide for the design and construction of breakwaters	BS 6349-7
Flat-bottomed, vertical, cylindrical storage tanks for low temperature service, Part 3: Recommendations for the design and construction of prestressed and reinforced concrete tanks and tank foundations, and for the design and installation of tank insulation, tank liners and tank coatings	BS 7777-3
Code of practice for earth retaining structures	BS 8002
Code of practice for foundations	BS 8004
Guide to design and construction of outfall	BS 8005
Code of practice for pipelines, pipelines subsea, Part 3: Design, construction and installation	BS 8010-3
Code of practice for ground anchorages	BS 8081

Issued by:
British Standards Institution
389 Chiswick High Road
London W4 4AL
UK

INTERNATIONAL STANDARDS

Quality assurance requirements for measuring
equipment,
Part 1: Metrological confirmation system for
measuring equipment

ISO 10012-1

*Issued by:
International Organisation for Standardisation
1, Rue de Varembe
CH-1211 Geneva 20
Switzerland*

*Copies can also be obtained from national standards
organizations.*

NOTE: Most referenced external standards are available for Shell users on the Shell Wide Web at address
<http://swwww.shell.com/standards>

14. BIBLIOGRAPHY

AMERICAN STANDARDS		
Water (I)	ASTM Vol. 11.01	Detailed standards for water and environmental testing
Water (II)	ASTM Vol. 11.02	Detailed standards for water and environmental testing
Recommended practice for design, construction, operation, and maintenance of offshore hydrocarbon pipelines	API RP 1111	Includes general guide on requirements for site investigation
<i>Issued by:</i> <i>American Society for Testing and Materials</i> <i>1916 Race Street,</i> <i>Philadelphia Pa 19103</i> <i>USA</i>		
BRITISH STANDARD		
Guide to geophysical logging of boreholes for hydrogeological purposes	BS 7022	Contains some guidance on requirements for site investigation
<i>Issued by:</i> <i>British Standards Institution</i> <i>389 Chiswick High Road</i> <i>London W4 4AL</i> <i>UK</i>		
NETHERLANDS STANDARD		
Geotechnics TGB 1990 - Basic requirements and loads	NEN 6740 Nederlands Normalisatie-Instituut	Basic requirements, includes methods of calculating design parameter values from measured data
<i>Issued by:</i> <i>NNI</i> <i>Kalfjeslaan 2,</i> <i>2600 GB Delft</i> <i>The Netherlands</i>		
EUROPEAN STANDARDS		
Geotechnical design, CEN Eurocode 7: Part 1: General rules	ENV 1997-1	Comprehensive standard; includes general guidance about site investigation approach and data interpretation
Geotechnical design, CEN Eurocode 7: Part 2: Design assisted by laboratory testing	ENV 1997-2	Comprehensive standard; includes detailed guidance about planning and requirements for geotechnical laboratory tests and data interpretation
Geotechnical design, CEN Eurocode 7: Part 3: Design assisted by field testing, general rules	ENV 1997-3	Comprehensive standard; includes detailed guidance about planning and requirements for geotechnical in-situ tests and data interpretation

Design provisions for earthquake resistance of structures, CEN Eurocode 8: Part 1-1: General rules - Seismic actions and general requirements for structures	ENV 1998-1-1	Comprehensive standard; includes general guidance for site investigation in seismic regions
Design Provisions for Earthquake Resistance of Structures, CEN Eurocode 8: Part 5: Foundations, Retaining Structures and Geotechnical Aspects	ENV 1998-5	Comprehensive standard; includes general guidance for site investigation in seismic regions
Aggregates for concrete including those for use in roads and pavements	prEN 12620	Listing of detailed requirements for quality and grading of aggregates obtained by processing natural materials for use in concrete
<i>Issued by:</i> CEN Secrétariat Central Rue de Stassart 36, B-1050 Brussels, Belgium <i>Copies can also be obtained from national standards organizations</i>		

OTHER DOCUMENTS

ASCE (1997) "Guidelines for Seismic Evaluation and Design of Petrochemical Facilities"

API RP 2A "Recommended practice for planning, designing and constructing fixed offshore platforms

Clayton C.R.I., Matthews M.C., Simons N.E. (1995), "Site Investigation", Second Edition, Blackwell Science

Construction Industry Research and Information Association CIRIA and Centre for Civil Engineering Research and Codes CUR (1991), "Manual on the Use of Rock in Coastal and Shoreline Engineering", CIRIA Special Publication 83 and CUR Report 154

Decourt, L., (1989), "General Report/Discussion Session 2: SPT, CPT, Pressuremeter Testing and Recent Developments in In-Situ Testing - Part 2: The Standard Penetration Test, State-of-the-Art Report", Proc. 12th Int. Conf. on Soil Mechanics and Foundation Engineering, Rio de Janeiro, Vol. 4, pp. 2405-2416.

Geological Society Engineering Group Working Party (1972), "The Preparation of Maps and Plans in terms of Engineering Geology", Quarterly Journal of Engineering Geology, Vol. 5, No. 4, pp. 295-382.

Geological Society Engineering Group Working Party (1988), "Engineering Geophysics", Quarterly Journal of Engineering Geology, Volume 21, pp. 207-271.

Hodgson A.J., Adam C.H. and Sneddon M. (1995), "Recent Developments in Planning and Execution of Nearshore Site Investigation", Proceedings of the International Conference on Advances in Site Investigation Practice, Institution of Civil Engineers, pp. 12-24

COMMENTS

Guidance on seismic hazard and design

Includes soil softening criteria under cyclic loads

Comprehensive guidance on geotechnical and geophysical surveys

Comprehensive guide on the use of rock for erosion control

Background information about Standard Penetration Test

Explanation and use of engineering geological mapping techniques

Background information on geophysical survey techniques

Considerations about planning and conducting site investigation surveys from vessels and jack-up platforms

Engineers, pp. 13-24.

International Society for Rock Mechanics ISRM (1981), "Rock Characterisation Testing and Monitoring: ISRM Suggested Methods", Edited by E.T. Brown, Pergamon Press England, 211 pp.

Lunne, T., Robertson, P.K. and Powell, J. (1977), "Cone Penetration Testing in Geotechnical Practice", Blackie Academic, 265 pp.

SIOP report OP 96-30326, July 1996, ORTEC/2 "Departmental Manual" ; "Recommended practice for environmental site assessment"

The Institution of Civil Engineers, ICE (1991), "Inadequate Site Investigation", Report by the Ground Board of the Institution of Civil Engineers on inadequate site and ground investigation leading to construction delays and additional costs, 26 pp.

Site Investigation Steering Group, SISG (1993) "Site Investigation in Construction, 2. Planning, procurement and quality management", Thomas Telford, London.

Site Investigation Steering Group (1993), "Site Investigation in Construction, 3. Specification for Ground Investigation", Thomas Telford, London.

Union Francaise des Geologues (UFG) Paris (1992) "Applied geophysics, code of practice". (code de bonne pratique) Contributing authors were BRGM, CGG, CPGF, LCPC.

Suggested methods for rock testing

Comprehensive guide on application and interpretation of cone penetration tests

Advice on environmental site investigations, for contaminated land, base line surveys etc.

Informative brochure-type document

Detailed guidelines for site investigation arrangements

Specification document for using geotechnical Survey Contractors with limited geotechnical expertise

Guidance on geophysical testing

APPENDIX 1 NOTES FOR GUIDANCE ON PROCUREMENT OF GROUND INVESTIGATIONS

1. Contents

1. Objective
2. Area of applicability of this document
3. Form of document
4. Objective of the ground investigation
5. Define amount of investigation required
6. Design of ground investigations
7. Contract documents
8. General advice

Record sheets for basic information required

Flow charts for procedures

2. Objectives

To quickly and efficiently guide the user to the selection of an appropriate investigation, and leave a file record of the reasons for that selection.

3. Area of Applicability of this Document

For use by all those concerned in the designing, specifying, or procuring ground investigations, or advising a client on any of the above.

4. Form of Document

The document comprises 3 parts:

1. A text section, setting out the steps required in the procurement of ground investigations.
2. A set of flow charts illustrating the overall process and details of various stages.
3. Check sheets/record sheets for defining the objectives and types of investigation proposed. The checklists, charts etc. are designed to be quickly and easily filled in to give a maximum of useful information for a minimum time input.

5. Objective of the Ground Investigation

For any successful project or piece of work, it is vital to define at the start what the objective is.

The objective of a ground investigation will normally be one of the following:

1. To select a suitable site for a project.
2. To prove whether or not a given site is suitable (feasibility study).
3. To obtain geotechnical design parameters and/or groundwater data for permanent or temporary works.
4. To assess contamination of a site.
5. To assess quantity, quality and availability of construction materials (e.g. sand or rock).
6. To investigate types of material to be excavated or dredged.

As sub-sections of (3) the following may apply

- To assess stability, drainage, settlement etc., effects of a new development or associated construction works on an existing site.

- To provide data for the comparison of alternative construction methods at the same or alternative sites.
- To investigate stability of existing structures, earthworks or natural features.
- To investigate foundation or ground failures or unexpected behaviour.

The objective should be filled in on record sheet I.

Note: If it is impossible to fill in the record sheet properly, there is insufficient information to proceed. Consult a geotechnical specialist, the project engineer or the client, as any further work is likely to be an ineffective use of time and money.

6. Define Amount of Investigation Required

1. This requires consideration of:

- Minimum information required, and preferred information to be obtained.
- Available information.
- Cost/budget.
- Is the ground expected to be good or bad?
- Is the ground expected to be uniform or variable?
- Method of payment for all parts of the contract.
- Acceptable risk (of problems or variations) during construction.
- Cost of returning for a second stage investigation (time and money).
- Complexity of geotechnical engineering (foundations/retaining structures, earthworks, etc.).
- How stringent are the design rules for settlement, ground movement, etc.?
- Complexity of ground water regime.

The tables on record sheet 3 allow for a simplified assessment of these factors, but should be used with care as some factors will require detailed investigation (for example the presence of a landslip or sinkhole on the site) irrespective of how good the "scores" are for other factors.

2. In assessing the extent of work required, a proper balance of field and laboratory work is required. For example, to spend USD 500,000 on offshore boreholes and then USD 2,000 on laboratory testing is unlikely to be a good balance.

The relative costs of different parts of the work should be considered, especially for offshore work where mobilisation and site costs are very high. For instance, replacement of one USD 25,000 borehole by extensive effective stress laboratory testing, or by full time site supervision and improved time on analysis of results, may give better value for money.

3. During detailed design, allowance will be made for uncertainties in the ground and ground parameters. These take the form of conservatism in design, i.e. extra costs to construct. A more detailed ground investigation will frequently reduce the overall project costs. The design conservatism can be formalised as in NEN 6740, or treated statistically, or based on experience.

7. Design of Ground Investigations

1. Ground investigations need to be designed if they are to achieve their objective. Design should be carried out by an individual or team with adequate training and experience.

8. Contract Documents

The following documents are required:

- specification (long or short);
- conditions of contract;
- instructions to tenderers;
- preamble to the bill of quantities;
- bill of quantities;
- form of agreement;
- programme;
- form of tender;
- contractor's offer and letter of acceptance.

9. General advice

1. Opting for the minimum possible investigation is not normally an optimal solution. What should be sought is an investigation giving the lowest total project cost. In estimating the costs the increased risk of a smaller investigation should be allowed for. Consider: Who pays? Who benefits?
2. Programme: It is difficult for the programme for a design and construct project to allow sufficient time for a ground investigation. Experience shows the following times to be around the shortest possible. Average times are longer, and actual times may be much longer in some cases.

Item	Minimum time (weeks)		
	Onshore (A)	Onshore (B)	Offshore (C)
Design GI, produce and approve documents	1	2	3
Tender period (from instruction to issue)	1	2	3
Evaluate tenders, negotiate and sign contract	0.5	2	3
Mobilise	0.5	2	3-6
Site works	varies (1+)	varies (2+)	varies (3+)
Preliminary report on field work	0.5	1	1
Laboratory testing (to preliminary report)	1-3	2-4	2-4
Final factual report	1	3	3
Interpretative report	1-3	1-3	2-10
TOTAL	6-10+	17-21+	23-36+

A = Country such as UK or NL, onshore

B = Less developed country*, onshore

C = Less developed country*, over-water

Timescales are normally longer in less developed countries, as, in general, data is less available, access is slower and plant availability is worse.

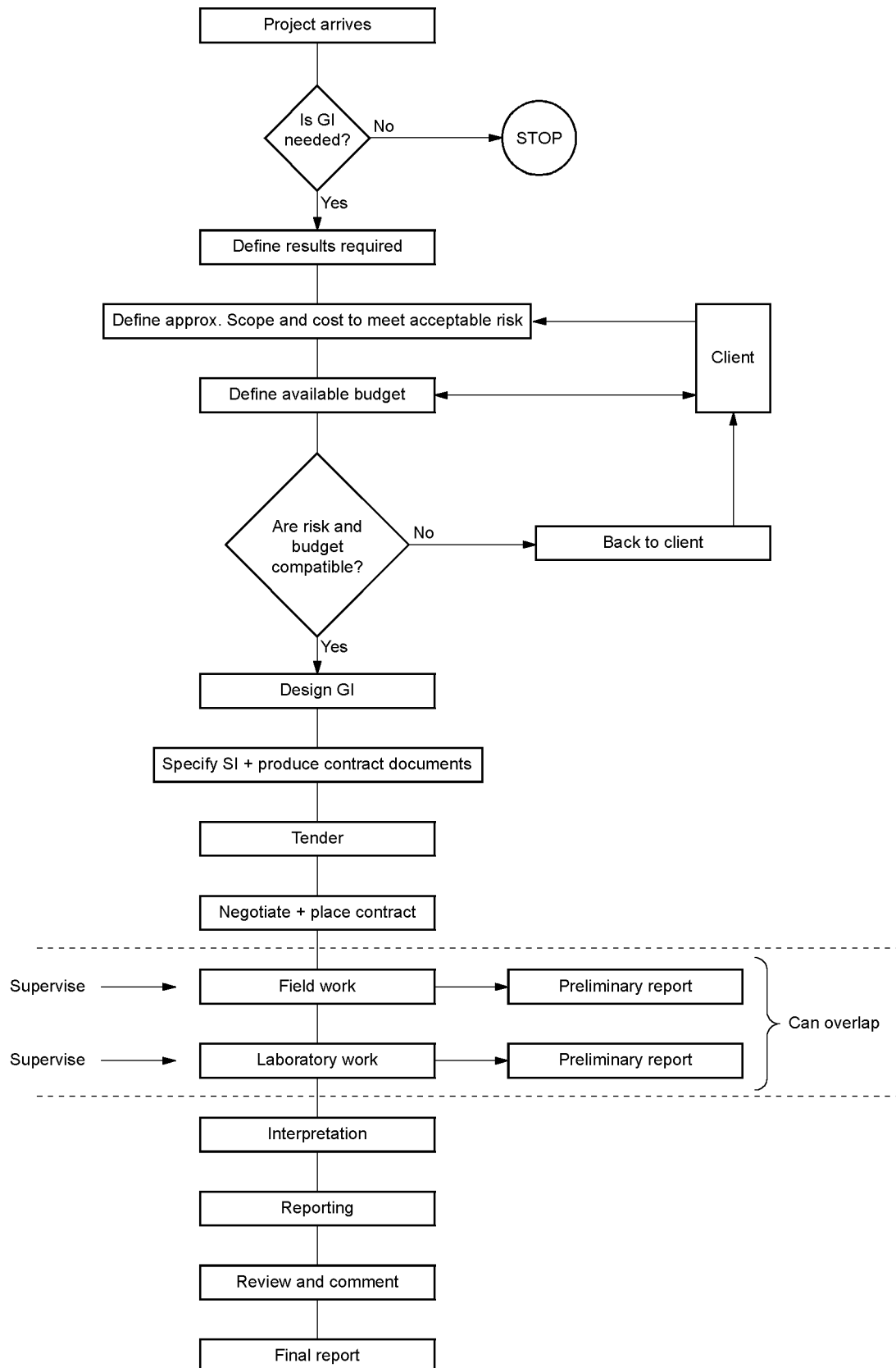
It can be seen that long timescales are required. Time savings are vital and can be made by:

- Carrying out (detailed) investigations at an earlier project stage than "required".

- Prior to award of main contract, agreeing the scope of investigations and inviting investigation subcontractors to tender.
- Good interim reporting. This is greatly improved by having a site representative/supervising engineer.
- Properly thought out co-ordination with the design process to avoid having to start the design process anew once final geotechnical information is available.
- Programming the field work to get right information first.

In order to ensure design data is sufficient, a supervising engineer should have the authority to change investigations to take into account unexpected results.

10 FLOW CHART FOR CORRECT PROCUREMENT OF GROUND INVESTIGATIONS



11. ATTACHMENTS

Record sheets

ATTACHMENT 1 GENERAL INFORMATION AND OBJECTIVES, 2 SHEETS

ATTACHMENT 2 EXTENT OF INVESTIGATION REQUIRED, 1 SHEET

ATTACHMENT 3 DERIVATION OF RISK FACTORS, 2 SHEETS

ATTACHMENT 4 DATA FROM WALKOVER SURVEY / SITE VISIT, 3 SHEETS

Record sheet 1, sheet 1 of 3

ATTACHMENT 1 GENERAL INFORMATION AND OBJECTIVES

Project Administration:

Project number:
Project name:
Project location:
Project engineer:
Client:

Brief project description:

.....
.....
.....

Programme:

Weeks available

Production of documents	
Tender + mobilisation	
Site Work	
Laboratory work	
Factual reporting	
Analysis and interpretation reporting	
Further investigations	

Budget:

Total

Production of documents, tending evaluation etc.	
Office based supervision	
Reporting and client liaison	
Direct costs (GI Contractor)	

Type of Contract:

Who supervises
Who sponsors
Legal system

Record sheet 1, sheet 2 of 3

Preferred / available contractors:

.....

.....

.....

.....

.....

General objectives of Investigation:

.....

.....

.....

Specific information required:

.....

.....

.....

Is site visit / walkover survey planned?	Yes/no
If yes, by geotechnical / non-geotechnical personnel?	Geotechnical / non-geotechnical
Likely availability of desk study information	None/poor/fair/good/excellent

Access:

.....

.....

Available resources (e.g. water, accommodation, power, sewerage, etc.):

.....

.....

Terrain:

.....

Record sheet 1, sheet 3 of 3

Previous site use (e.g. greenfield, previous industry) and **potential associated hazards** (e.g. contamination, voids, obstructions):

.....
.....

Water depth, likely waves or currents (if over water):

.....
.....

Contact names and addresses:

.....
.....
.....
.....
.....
.....
.....

Filled in by: Date:

Record sheet 2 : sheet 1 of 1

ATTACHMENT 2 EXTENT OF INVESTIGATION REQUIRED

Scope

Preliminary / detailed design / investigation of failure / other (state.....)

Anticipated

Site visit.	Yes/no	If no, why not
Desk study	Yes/no	If no, why not
Single stage site investigation	Yes/no	Phased? Yes/no
Multi-stage site investigation	Yes/no	
Provision of on site supervision	Yes/no	

Is investigation onshore and/or tidal/shallow water and/or offshore?

Special investigations needed for environmental / contamination issues Yes/no

Available information

None / poor / fair / good / excellent

Acceptable risk from geotechnical aspects during design and construction (including effects on programme)

Extremely low / reasonable / very flexible

Complexity of geotechnical design / construction

Low / medium / high

Percentage of main contract that is geotechnical related work

Low / medium / high

Fixed Budget Restrictions

Severe / normal / open budget / unclear

Ground conditions anticipated

Poor / fair / good

Cost (including delay) of re-mobilising for second stage

Investigation if first stage is inadequate

Cheap / moderate / expensive Sum (if known).

Contractor anticipated

Experienced / less experienced. With / without supervision

Filled in by: Date:

Record sheet 3, sheet 1 of 2

ATTACHMENT 3 DERIVATION OF RISK FACTORS

Sheet 1: BASIC FACTORS

Fill in for the proposed site and project stage. For large projects it is convenient to consider the following three stages: predefinition (feasibility & scouting); pre-EPC contract (BOD, BDP, BDEP); and EPC contract.

Site

Project Stage

	Description	Score		
		1	2	3
	Ground			
	A Anticipated ground quality	Poor	Fair	Good
	B Geological complexity of site area	High	Medium	Low
	Ground score = (2x score A) + Score B 8,9 = Good 5,6,7= Fair 3,4 = Poor (Ring score)			
	Structure of proposed construction			
	C Structure complexity	High	Medium	Low
	D State of definition of structure	Poor	Fair	Good
	E Structure sensitivity to geotechnical factors	High	Medium	Low
	Structure score - score (C + D + E) 8,9 = Good 5,6,7= Fair 3,4 = Poor			
	Information			
	F Available geotechnical information	Poor	Fair	Good
	G Meaningful geotechnical site visit before GI?	No	Unknown	Yes
	H Meaningful desk study before GI?	No	Unknown	Yes
	Information score = (2xF) + G + H 10,11,12 = Good 7,8,9 = Fair 4,5,6 = Poor			
	Risk			
	Acceptable geotechnical risk	Low	Medium	High
	Sensitivity (costs, programme) to geotechnical risk	High	Medium	Low
	Percentage of works that is geo-related	High	Medium	Low
	Risk score = (I + J + K) 8,9 = Good 5,6,7 = Fair 3,4 = Poor			

Record sheet 3, sheet 1 of 2

DERIVATION OF RISK FACTORS

Sheet 2 : COMBINED FACTORS

Combined ground + structure factor (G+S)

Ground	Good	3	4	5
	Fair	2	3	4
	Poor	1	2	3
		Poor	Fair	Good
		Structure		

Combined risk and available data factor (R+D)

Risk	Good	3	4	5
	Fair	2	3	4
	Poor	1	2	3
		Poor	Fair	Good
		Data		

Overall risk/degree of investigation guide

G + S	5	C	C	B	A	A
	4	C	C	C	B	A
	3	D	C	C	C	B
	2	E	D	C	C	C
	1	E	E	D	C	C
		1	2	3	4	5
		R + D				

A = Low risk, low degree of GI needed

E = Highest risk, highest degree of GI needed, e.g. staged investigation and/or special techniques, monitoring etc. required.

Preliminary estimate of type of investigation based on overall risk

Risk factor	Investigation
A	Single stage, fully pre-determined.
B	Single stage, some site flexibility.
C	Probably single state, but priority for some boreholes/CPTs. Flexibility required on site to add/vary exploratory hole locations.
D	Single or two stage investigation. Some provisional extra BH/CPTs. Testing to be allowed for in BoQ and specified on site.
E	Staged investigation. Results of first phase used to specify second. Special testing or instrumentation likely (note: stages can run consecutively within a single contract if evaluation of results is rapid enough).

Filled in by: Date:

Walkover survey record sheet 1 of 5

ATTACHMENT 4 DATA FROM PRELIMINARY SITE VISIT

(Also referred to as WALKOVER SURVEY)

1. What is the topography of the site, i.e. flat, sloping, undulating? Describe in detail.

.....
.....

2. Previous uses of land?
i.e. unused, farming (type), industrial, domestic, etc.

.....

3. Vegetation. Is site cleared; if so, what was previous growth? Describe trees, scrub, existing vegetation and density of growth.

.....
.....

4. Is contamination likely? Yes / No

5. Is Environmental Site Assessment recommended e.g. for
Assessing liabilities prior to land purchase Yes / No
Baseline survey Yes / No

Other Yes/No Reason

6. Adjoining land use. Topography and usage i.e. farming, industrial, mining or quarrying, tipping, etc.

.....

7. Is the site on undisturbed ground? (Refer to questions 1, 2 and 4.) (If no give details)

.....

8. Is the site or adjoining area likely to be subject to subsidence and/or landslip? (Refer to questions 1 and 5)

.....

9. Do the maps and other information show uniform geological conditions? (If no give details)

.....

10. Is the site liable to flooding? (If yes give details).

.....

Walkover survey record sheet 2 of 5

11. Is there a high seasonal water table, evidence of springs or land drains?

.....

12. What is the predominant soil type?

- Clay (soft/stiff)
- Silt
- Sand
- Gravel
- Rock
- Other (specify)
- Is it an earthquake zone? If yes, give details.

.....

.....

Walkover survey record sheet 3 of 5

1. What problems are likely to be associated with the ground? Refer to list of problem ground conditions in Appendix 3.

.....

2. Is the site likely to be difficult to work in particular seasons? (If yes, why?)

.....

3. Are there existing foundations on the site? Make comments on extent of these and possible implications.

.....

.....

4. Is there evidence of pipelines or services within or in close proximity to the site? If yes, give details.

.....

.....

5. Are the following likely to be present?
a) Restrictions to the developable area?
b) Exceptional costs in development?

6. Are there any trees on the site adjoining land likely to affect foundations, sewers or roads due to heave or shrinkage (guidance notes available from Building Research Establishment in UK). If so, what are their height, girth, species, spacing, and occurrence (mark on plan).

.....

7. Comment on the likelihood that normal shallow footings will be adequate.

.....

.....

8. If not:

- a) Why not?
b) What type of foundations are likely to be required?

.....

.....

9. Are there likely to be any special provisions for the foundations with regard to depth, reinforcement, width?

.....

.....

.....

Walkover survey record sheet 4 of 5

10. Are there visible obstructions to piling (e.g. rock outcrops, construction debris, old foundations)?

If yes give details:

.....
.....

11. Are there likely to be any special provisions for the ground floor? Suspended, reinforced, etc.?

.....

12. Are there likely to be any special provisions for the structure?

.....

13. Are there likely to be any special provisions for the drainage? (e.g. piled in soft ground)

.....

Walkover survey record sheet 5 of 5

14. Other observations about the site gained during site visit/walkover. Include relevant aspects of access, marine facilities, synergies with other projects, third party interests, political aspects, etc.

This image shows a full page of primary-ruled notebook paper. It features multiple sets of horizontal lines designed to guide young learners' handwriting. Each set consists of three lines: a solid top line, a dashed middle line, and a solid bottom line. These sets are repeated down the entire page, providing ample space for practicing letter formation and alignment. The paper is otherwise blank, with no text or other markings.

Filled in by: Date:

APPENDIX 2 COST ASSESSMENT FOR NEARSHORE GEOTECHNICAL SURVEY

The following table illustrates Survey Contractor costs for a high-standard nearshore geotechnical survey relative to costs for an on-land survey.

The cost assessment considers the following aspects:

- technical standards set by this DEP are applicable;
- Shell HSE standards are applicable, in particular to nearshore activities;
- total cost includes 15 boreholes to 25 m depth, laboratory testing and factual reporting, but excludes costs incurred by Principal or Contractor for survey commissioning and management of the survey contract;
- mob cost comprises mobilisation and demobilisation of equipment, personnel and samples for a project site at 1000 km distance from the home base of the Survey Contractor;
- operational nearshore activities include a small jack-up platform for 0 m to 10 m water depth, a large jack-up platform or drill ship for 10 m to 25 m water depth and small support vessels;
- negligible weather down-time applies to nearshore activities.

APPENDIX 3 GROUND AS POTENTIAL HAZARD

DESCRIPTION	POTENTIAL HAZARD
GROUND TYPE	
unsaturated soil	loss of strength and stiffness upon wetting; collapse upon wetting
frozen soil	loss of strength and stiffness upon thawing; volume change
gassy soil	high compressibility, low strength; explosive; flammable; toxic
man-made fill	high compressibility, low strength; non-homogeneous; foreign objects/obstructions; degradable material; contamination; voids
brownfield sites	foreign objects/obstructions; degradable material; contamination; voids
saline alkali soil (incl. Sabkha)	volume change; very aggressive towards concrete and steel, hard crust, soft underneath
ferrous soil	cemented (iron) layers, clogging of wells
organic soil	high compressibility, large frost heave
peat	high compressibility, large frost heave
dispersive soil	internal erosion upon flow of water through soil; formation of cavities
volcanic soil	instability of soil structure upon disturbance, low density
loess	instability of soil structure upon disturbance
till, boulder clay and glacial soil	occurrence of cobbles and boulders, non-homogeneous
shrinking/swelling clay and silt (or active clay)	loss of strength and stiffness upon increase in water content; change in volume upon change in water content
(heavily) overconsolidated clay	swelling and loss of strength in excavations; high lateral ground pressure on retaining walls, pits, etc.; loss of strength under cyclic loading; often high undrained strength but low drained strength
(stiff) fissured clays	(undrained) strength of small intact sample greater than insitu strength
normally consolidated and slightly overconsolidated clay and silt	high compressibility, low strength
underconsolidated clay and silt	ongoing compression under self-weight, negative skin friction on piles

DESCRIPTION	POTENTIAL HAZARD
sensitive clay (or quick clay)	loss of strength upon application of shear strain, change of groundwater from salt to fresh or disturbance (can turn into liquid)
fissured clay	low strength; high permeability
varved clay (or laminated or stratified clay)	anisotropy in strength, stiffness and permeability
calcareous clay and silt	cemented layers; soluble
sand	scour and soil movement upon application of water flow
loose sand and silt	static and cyclic liquefaction
mica sand	instability of soil structure upon disturbance; low shear strength; low pile skin friction
glauconitic sand	weak particles, "clay" behaviour, low permeability
calcareous (carbonate) sand	low strength and stiffness upon application of stress; cemented layers; soluble; compressible particles; low driven pile skin friction
volcanic sands	light chemical cementing; low driven pile skin friction
chalk	solution holes; mining activities; collapse settlement of fill
basalt	often contains minerals deleterious in concrete aggregates
commercially valuable deposits	mining activities
coral	crushable, vary variable strength, voids
residual soil, weathered granite, etc.	strength, compressibility and permeability strongly depend on soil structure; non-homogeneous differential weathering; occurrence of boulders
claystone and shale	swelling upon reduction in stress
evaporitic rock	soluble; mining activities
gypsum and anhydrite	volume change
coal	flammable; mining activities
acid soils	a number of soils become highly acidic when exposed to air (or aerobic groundwater). pH can drop as low as 2. Particularly prevalent where sulphides e.g. pyrites are present.
strong rock or better	rock burst upon reduction in stress

DESCRIPTION	POTENTIAL HAZARD
STRUCTURAL FEATURE	
oil or gas extraction	surface subsidence
mining activities	surface subsidence; cavity collapse; high permeability zone
artesian groundwater	ground instability; flooding
land slide	re-activation because of low strength of failure surface; disturbed non-homogeneous ground; interrupted groundwater regime
cliffs	instability, rockfalls
caprock	often has voids immediately beneath
scree	marginal stability; impossible to drill through
(steeply) dipping rock	driven piles. CPTs and boreholes tend to bend to follow bedding planes, then break
bedding plane shear	very low shear strength discontinuity in rock
claystone, siltstone, sandstone sequence	differential weathering
dyke and sill	high or low strength and stiffness zone; high or low permeability discontinuity
karst (eroded limestone)	voids, collapse of cavities; sinkhole formation; high permeability zones, irregular buried rock surfaces
fault	low strength zone; high or low permeability discontinuity; ongoing activity
volcano	ash and lava outburst; tremors; gas emission

APPENDIX 4 SAMPLING TECHNIQUES AND SAMPLE CLASS

SAMPLE CLASS		
Designation	Sample Disturbance	Laboratory Test Feasibility
Class 1	undisturbed	strength, stiffness and consolidation
Class 2	undisturbed	layering, permeability, unit weight
Class 3	disturbed	water content
Class 4	disturbed	particle size analysis, Atterberg limits, soil type
Class 5	disturbed	stratigraphy

Notes Adapted from BS 5930
The higher class includes the laboratory tests of the lower class.

SAMPLING TECHNIQUE AND SAMPLE CLASS

Sampling Technique	Inner Diameter D_i (mm)	Sample Length	Sampler Insertion Method	Ground Type Feasibility	Sample Class
thin-walled open-tube sampler	50 to 100	$2.5 D_i$ to $10 D_i$	hammer or push	very soft cohesive soil	2 (1)
				soft to stiff cohesive soil	1
				very stiff cohesive soil	2 (1)
				cohesionless soil above ground water level	3 (2)
thin-walled open-tube sampler	35 to 50	$5 D_i$ to $20 D_i$	hammer or push	cohesive soil	3 (2)
				cohesionless soil above ground water level	4
thick-walled open-tube sampler	50 to 100	$2.5 D_i$ to $10 D_i$	hammer or push	soft to hard cohesive soil	2 (1)
				cohesionless soil above ground water level	3 (2)
				cohesionless soil below ground water level	4 (3)
thick-walled open-tube sampler, including split-barrel SPT	35 to 50	$5 D_i$ to $20 D_i$	hammer or push	most soil types	4 (3)

SAMPLING TECHNIQUE AND SAMPLE CLASS					
thin-walled piston sampler	50 to 100	$\leq 10 D_i$	push	very soft to firm cohesive soil	1
				loose to medium dense cohesionless soil above ground water level	3 (2)
block sampling	as required	as required	cutting from (excavation) exposure	soft to hard cohesive soil	1
				rock	1
core barrel	50 to 100	1 m to 3 m	rotary drilling	very stiff to hard cohesive soil	2 (1)
				rock	1
vibrocore sampler	50 to 100	1 m to 6 m	self-weight and high-frequency vibration	very soft to soft cohesive soil	3 (2)
				firm to hard cohesive soil	2 (1)
				cohesionless soil below ground water level	4 (3)
gravity core sampler	50 to 100	1 m to 6 m	self-weight	soft to hard cohesive soil	2 (1)
				cohesionless soil below ground water level	4 (3)
sampling from auger or cable percussion tool	not applicable	continuous	drilling	most soil types above ground water level	3
				cohesive soil below ground water level	4
				cohesionless soil below ground water level	5
sampling of cuttings from rotary drilling or wash boring	not applicable	continuous	drilling	most soil and rock types	5
				cohesionless soil below ground water level	5

- Note:
1. Adapted from BS 5930.
 2. Sample class in brackets may be feasible under favourable conditions.
 3. Sampling of cohesionless soil may require the use of sample retainer basket (core catcher).

APPENDIX 5 SOIL DESCRIPTION AND CLASSIFICATION

The BS 5930 and ASTM D2487 systems for (silica-based) soil description and classification are in common use for international projects. They are broadly equivalent as they are both based on Casagrande's Unified Classification System, but they are not the same. Karim and De Ruijter list some differences:

- Fine grained soil according to BS 5930 consists of particles smaller than 60 mm and the percentage by dry weight passing the 0.06 mm sieve exceeds 35%. ASTM D2487 considers 75 mm, 0.075 mm and 50% boundaries respectively.
- ASTM defines silty clays on the plasticity chart in a zone bounded by plasticity indices of 4% and 7% and Casagrande's A-line. BS considers a border formed by a plasticity index of 6% and Casagrande's A-line.
- Other differences between BS and ASTM include the use of group symbols, main and secondary soil fractions, terminology and test procedures.

Calcareous sediments present a further major challenge. Separate soil description systems are not uncommon. Peuchen et al. (1998) present a single description system on the basis of ASTM and BS description, but expanded with the Clark and Walker (1977) description system for calcareous soil and soft rock.

References:

Clark A.R. and Walker B.F. (1977), "A Proposed Scheme for the Classification and Nomenclature for Use in the Engineering description of Middle Eastern Sedimentary Rocks", *Geotechnique*, Vol. 27, pp. 94-99.

Karim U.F. and De Ruijter M.R. (1993), "A Measurable Classification System for Non-Calcareous Marine Soils", in: *Offshore Site Investigation and Foundation Behaviour*, Society for Underwater Technology, Arhus, D.A. et al., eds., Vol. 28, pp. 53-76.

Peuchen J., De Ruijter M. and Goedemoed S. (1998), "Commercial Characterisation of Calcareous Soils", *Second Int. Conf. on Engineering for Calcareous Sediments*, Bahrain (1999).

APPENDIX 6 GEOTECHNICAL FIELD SURVEY TECHNIQUES

Survey Technique	Main Application
VISUAL INSPECTION AND SAMPLING	
Engineering Geological Mapping	Inspection of ground surface conditions.
Trial Pit, Shaft, Adit	Inspection of exposed ground conditions; opportunity for sampling and in-situ testing of soil and rock.
Auger Borehole	Cutting samples available, opportunity for sampling and in-situ testing of soil only.
Cable Percussion Borehole	Cutting samples available, opportunity for sampling and in-situ testing of soil only.
Open-hole Rotary Borehole	Cutting samples available, opportunity for sampling and in-situ testing of soil and rock.
Rotary Core Borehole	Continuous sampling, opportunity for in-situ testing of stiff soil and rock.
Open drive samplers	Continuous sampling, open sided driven tubes. Small diameter disturbed sample, useful for logging strata. Can be hand carried.
Vibrocore and Gravity Core	Continuous sampling of soil to limited depth below ground surface (offshore and nearshore).
Tube Sampler	Discontinuous sampling of soil in borehole.
Water Sampler	Discontinuous sampling of water in borehole or piezometer.
Pore Water Sampler	Push-in probe for sampling of pore water in soil.
Video Logging	Visual inspection of borehole side.
CONTINUOUS IN-SITU TESTS	
Diagraphy	Continuous strength/stiffness profiling by monitoring borehole drilling behaviour in soil and rock.
Dynamic Probing (DP)	Continuous strength/stiffness profiling by dynamic penetration of probe into soil and soft rock. Various versions, typically from SPT hammer weight and drop to small hand carried.
Cone Penetration Test (CPT)	Continuous strength/stiffness profiling by push-in penetration of probe into soil and soft rock. An increasing number of more sophisticated CPT probes are being developed, able to improve information recovery by measuring other parameters.
Electrical Conductivity Probe	Continuous electrical resistivity profiling by push-in penetration of probe into soil and soft rock, often in combination with CPT.

Survey Technique	Main Application
DISCONTINUOUS IN-SITU TESTS	
Standard Penetration Test (SPT)	Combined open-tube sampling and in-situ testing by dynamic penetration of soil and soft rock.
Piezo-cone Dissipation Test	In-situ coefficient of consolidation for cohesive soil, in combination with CPT.
Rock Dilatometer Test (RDT)	In-situ stiffness of rock in borehole.
Pressuremeter Test (PMT)	In-situ stiffness/strength of soil and soft rock in borehole or from limited push-in penetration.
Cone Pressuremeter Test (CPMT)	Continuous CPT combined with PMT in soil.
Flat Dilatometer Test (DMT)	In-situ stiffness/strength of soil in borehole or from push-in penetration.
Field Vane Test (FVT) (often referred to as Shear Vane Test)	In-situ test for undrained shear strength of cohesive soil in borehole or from limited push-in penetration.
Hand shear vane (HSV)	Small, hand held version of the FVT, typically used in measuring soil strength in sides of trial pits, in ends of samples, and (in combination with extension rods) in early field studies and when only hand carried equipment is possible.
Plate Loading Test (PLT)	In-situ stiffness of soil and rock at ground surface or in borehole or in excavation.
In-Situ Density	In-situ density of soil and rock at ground surface or in excavation (various methods).
In-Situ CBR	In-situ strength/stiffness of soil and rock at ground surface or in excavation (for pavement design purposes).
Needle Probe	Thermal conductivity of soil and rock in borehole or by push-in penetration.
Hydraulic Fracturing	Hydraulic fracture pressure of soil and rock in borehole.
Borehole/Piezometer Permeability and Packer Test	Permeability of soil and rock by constant head, falling head and variable head water pressure variations.
Pumping Test	In-situ permeability of soil and rock by pumping in borehole and observation of drawdown in observation wells.
VARIOUS	
Geotechnical Instrumentation	Measurement of parameters such as deformation, stress, acceleration, water level, permeability, temperature etc.
Centrifuge Test	Small-scale model for assessment of ground/structure interaction.
Large-scale Test	Large-scale model for assessment of ground/structure interaction, including back-analysis.

Notes: 1. Downhole, surface-based, offshore, air-borne and satellite geophysical techniques may be part of a geotechnical survey. The table excludes such survey techniques.

2. Some geotechnical techniques overlap with environmental survey techniques.

APPENDIX 7 BOREHOLES AND SAMPLING

Borehole drilling:

- Drilling Equipment: any equipment that provides a suitably clean open hole before insertion of downhole sampling and/or testing apparatus and ensures that sampling and/or testing is performed in undisturbed ground.
- Drill Rig: machine capable of providing:
 - rotation, feed and retraction to drill pipe, casing and/or auger;
 - drill fluid pumping capacity, as required;
 - sampler or test apparatus insertion.
- Drill Casing: cylindrical pipe with one or more of the following purposes:
 - to support the sides of a borehole;
 - to support drill pipe above ground surface during over-water drilling;
 - to promote return of drilling fluid.
- Drill Pipe: cylindrical pipe connecting drill rig and drill bit.
- Drill Bit: device attached to drill pipe and used as a cutting tool to drill into the ground.

Core drilling:

- Single Tube Core Barrel: hollow steel tube with a head at the upper end threaded for drill pipe, and a threaded connection for the core bit at the lower end.
- Double Tube Core Barrel: assembly of two concentric steel tubes joined at the upper end by means of a swivel arranged to permit rotation of the outer tube without causing rotation of the inner tube; the upper end of the outer tube is threaded for drill pipe and the lower end is threaded for the core bit.
- Double Tube Core Barrel with Retrievable Inner Tube (Also referred to as triple tube core barrel): double tube core barrel that permits retrieval of the core-laden inner tube assembly to the surface through matching drill pipe without the need for withdrawal of the drill pipe.
- Core Bit: device attached to the core barrel and used as a cutting tool to drill into the ground.
- Core Catcher: device that assists retention of core in the core barrel.
- Core Box: box with longitudinal separators for the protection and storage of core.

Drive Sampling:

- Drive-Weight Assembly: Device consisting of hammer, hammer fall guide, anvil and hammer drop system.
- Hammer: impact mass which is successively lifted and dropped to provide the energy that accomplishes sampler penetration.
- Hammer Fall Guide: guide arrangement for the fall of the hammer.
- Anvil: drive-head which the hammer strikes and through which the hammer energy passes into the sampling rods.
- Hammer Drop System: pick-up and release mechanism by which lifting and dropping of the hammer is accomplished.
- Cathead: rotating drum in a rope-cathead hammer drop system around which a rope is wrapped to lift and drop the hammer by successively tightening and loosening the rope turns around the drum.

- Self-Tripping Release: hammer drop system that ensures a free fall of the hammer after lifting by a cable or rope.
- Free-Fall Winch: hammer drop system that permits a free release of the rotating drum of the winch around which a cable is wrapped to lift and drop the hammer.
- Sampling Rods: rods that connect the drive-weight assembly to the sampler head.

Push Sampling:

- Sampler Insertion Equipment: apparatus providing relatively rapid continuous penetration force.
- Reaction Equipment: reaction for the sampler insertion equipment.
- Sampling Rods: rods that connect the sampler insertion equipment to the sampler head.

Sampler:

- Open-Tube Sampler: sampler with tube that is open at one end and fitted to the sampler head at the other end.
- Piston Sampler: sampler with close fitting sliding piston that is held stationary during penetration of a flush sample tube into ground.
- Sampler Head: coupling between sampling rods and sample tube, and containing a non-return valve to allow free exit of water and air above sample.
- Sample Tube: cylindrical tube with cutting edge or cylindrical tube fitted with separate cutting shoe.
- Thin-Walled Sample Tube: sample tube with area ratio of less than 15 % and inside clearance ratio of less than 1 %.
- Thick-walled Sample Tube: sample tube not meeting the requirements of a thin-walled sample tube.
- Core Catcher: device that assists retention of the sample in the sample tube.

The definitions of area ratio and inside clearance ratio of a sampler are as follows:

Area Ratio: Indication of volume of ground displaced by the sample tube, calculated as follows:

$$A_r = [(D_e^2 - D_i^2) / D_i^2] \times 100$$

where:

A_r = area ratio expressed as percentage;

D_e = maximum external diameter of the sample tube and/or cutting shoe;

D_i = minimum internal diameter of the sample tube at the cutting edge;

Inside Clearance Ratio: Indication of clearance of sample inside the sample tube, calculated as follows:

$$C_r = [(D_t - D_i) / D_i] \times 100$$

where:

C_r = inside clearance ratio expressed as percentage;

D_t = internal diameter of the flush portion of the sample tube;

D_i = minimum internal diameter of the sample tube at the cutting edge.

APPENDIX 8 STANDARD PENETRATION TEST

Descriptions of common SPT apparatus are as follows:

- Drive-Weight Assembly: device consisting of hammer, hammer fall guide, anvil and hammer drop system.
- Hammer: impact mass of 63.5 kg which is successively lifted and dropped over 0.76 m to provide the energy that accomplishes the sampling and penetration.
- Hammer Fall Guide: guide arrangement for the fall of the hammer.
- Anvil: drive-head which the hammer strikes and through which the hammer energy passes into the sampling rods.
- Hammer Drop System: pick-up and release mechanism by which lifting and dropping of the hammer is accomplished.
- Cathead: rotating drum in a rope-cathead hammer drop system around which a rope is wrapped to lift and drop the hammer by successively tightening and loosening the rope turns around the drum.
- Self-Tripping Release: mechanism that ensures a free fall of the hammer.
- Sampling Rods: rods that connect the drive-weight assembly to the split barrel sampler or solid cone.
- Split Barrel Sampler: flush sampler with 35 mm ID (lined) or 38 mm ID (unlined) and 51 mm OD, of minimum 0.5 m length and with a longitudinal split.
- Split Liner: internal lining of the split barrel sampler.
- Core Catcher: device that assists retention of the sample in the split barrel sampler.
- Solid Cone: conical part having a 60° apex angle that replaces the drive shoe of the split barrel sampler under special conditions.

The procedure for a Standard Penetration Test includes the following:

- advancement of the borehole to the selected test depth, while maintaining the drill fluid at or above the in-situ groundwater level;
- lowering of the split barrel sampler and sampling rods into the borehole and positioning of the drive weight assembly;
- marking of the sampling rods in three successive 0.15 m increments;
- driving the split barrel sampler with the hammer and counting the number of blows applied in each increment until one of the following occurs:
 - the total number of blows reaches 100;
 - the number of blows for a 0.15 m increment reaches 50;
 - there is no observed advance of the split barrel sampler during the application of 10 successive blows;
 - the split barrel sampler completes the 0.45 m penetration without reaching a limiting blow count.
- retrieval of the sampler and removal of the sample for logging and storage;
- advancement of the borehole to the next test depth.

The initial 0.15 m penetration is the seating drive. The N-value or the standard penetration resistance is the sum of the number of blows required for the second and third 0.15 m increments. If the sampler penetration is less than 0.45 m then the results include:

- the number of blows per each complete increment;
- the number of blows per partial increment;

- the depth of penetration for the partial increment.

In this situation, it is common practice to apply linear extrapolation to a blow count for 300 mm penetration to obtain the "N-value". This extrapolation usually takes account of the blow count for the seating drive. The results can include the initial self-weight penetration of the split barrel sampler below the bottom of the borehole, if significant.

Geotechnical practice may require correction factors for comparative studies and/or to account for regional variations in practice. The more important correction factors are kinetic energy, "enthru" (delivered energy) and critical (rod) length. For example, the so-called N_{60} -value denotes an N-value corrected to 60% enthru energy.

SPT 'N' values have to be corrected for effects of depth/in-situ stresses before use in many design methods.

APPENDIX 9 ELECTRIC CONE PENETRATION TEST

Electric cone penetration test: measurement process whereby, from a certain point vertically downward, the cone resistance and, if applicable, the sleeve friction of the soil are measured and recorded relative to the achieved depth, using a cone penetrometer equipped with internal load sensors which is pushed into the soil at a constant rate of penetration.

Cone resistance: ratio of the force required to move the cone downward during cone penetration testing, to the cross-sectional area of the base of the cone.

Sleeve friction: ratio of the force required to move the friction sleeve downward during cone penetration testing, to the external cylindrical surface area of the friction sleeve.

Friction ratio: ratio of the sleeve friction measured at a particular depth, to the cone resistance measured at the same depth, expressed as a percentage.

CPT length: Sum of the lengths of the push rods and the cone penetrometer, reduced by the height of the conical part of the cone, relative to a fixed horizontal plane.

CPT depth: depth of the base of the cone, relative to a fixed horizontal plane.

Net area ratio: ratio of the cross-sectional area of the cone penetrometer above the cone and at the location of the groove, to the nominal cross-sectional area of the base of the cone.

Zero drift: absolute difference of the zero of the measuring system at the start of the cone penetration test and the zero at the completion of the cone penetration test.

Friction-cone penetrometer: cone with an associated internal load sensor and a friction sleeve with an associated internal load sensor and, if applicable, an internal inclinometer.

Piezo-cone penetrometer: as for friction-cone penetrometer, but also an internal water pressure sensor

Depth sensor: Apparatus required for measurement of the CPT length.

Heave compensator: Apparatus for selection of measurements if upward movements of the push rods occur relative to the depth sensor, as a result of a decrease in force on the push rods.

Push rods: A string of connected rods for transfer of a compressive or uplift force to the cone penetrometer.

Friction reducer: a local and symmetrical widening of the diameter of the push rod, for reduction of the friction along the push rods.

Push rod casing: Apparatus to prevent buckling of the push rods.

Thrust machine: Apparatus for application of vertical thrust to penetrate the cone penetrometer and the push rods into the soil at a constant rate of penetration.

APPENDIX 10 STANDPIPE PIEZOMETER

Casing: cylindrical pipe which is installed to isolate the zone being monitored.

Protective Casing: section of larger diameter pipe that is placed over the upper end of a smaller diameter riser or casing, to provide structural protection to the standpipe piezometer, to restrict unauthorised access to the riser, and to prevent surface water from entering the annulus.

Weep Hole: small diameter hole in the protective casing above the ground surface that serves as drain hole for water that may enter the protective casing annulus.

Vented Cap: cap with a small hole that is installed on top of the riser.

Riser: pipe extending from the well screen to or above the ground surface.

Well Screen: filtering device used to retain the filter pack; usually a cylindrical pipe with openings of a uniform width, orientation and spacing.

Sediment Sump: blank extension beneath the well screen used to collect fine-grained material from the primary filter pack and adjacent strata.

Primary Filter Pack: sand or sand/gravel mixture of selected grain size and gradation that is installed in the annular space between the borehole wall and the well screen, extending an appropriate distance above the well screen.

Secondary Filter Pack: sand or sand/gravel mixture of selected grain size and gradation that is installed in the annulus between the primary filter pack and the overlying annular sealant, or between the annular sealant and overlying grout backfill, or both.

Annular Sealant: material used to seal annular space between two concentric tubes or casings, or between casing and the borehole wall, including space(s) between multiple strings of tubing/casing.

Tamper: cylindrical section of tubing that slips over the riser and is used to tamp annular sealant or filter pack material into place.

Tremie Pipe: pipe or tube used to transport filter pack material and annular sealant from the ground surface into the annular space.

Centraliser: device that assists in centring a casing or riser within a borehole or another casing.

Standpipe piezometer development and water level monitoring may require apparatus such as a bailer or pump to withdraw water from a standpipe piezometer or borehole, and a water level measuring device.

APPENDIX 11 POINT LOAD STRENGTH INDEX

Point load strength index $Is_{(50)}$ provides information for strength classification of rock. Some rock classification systems include $Is_{(50)}$ as direct parameter, but most systems consider uni-axial compressive strength (σ_c or UCS) as strength parameter. Correlations between $Is_{(50)}$ and σ_c are common. Typical relationships are in the form $\sigma_c = k * Is_{(50)}$, where k ranges between 20 to 25, but the range may be as wide as $k = 4$ to $k = 50$. A value of $k = 4$ has been reported by Abbs (1985) for some soft carbonate rock types. A value of $k = 50$ may be indicative of significant rock anisotropy (ISRM, 1985). Soft rocks often show non-linear relationships or no statistically reliable relationship at all. Comments are as follows:

- Errors in measurement of point load strength are not uncommon, in particular because point load strength testing often takes place in the field, rather than in a controlled laboratory environment. Gross errors can include: calibration out-of-date, damage to apparatus and operator not familiar with details of test procedure. A common error is assuming linear calibration of force to apparatus gauge pressure for a situation where an offset value applies. This error can be significant for soft rock.
- Determination of uni-axial compressive strength takes place in a laboratory environment, which typically allows greater care during testing. However, a particular consideration is the accuracy achieved in specimen trimming (or machining). The ends of the specimen should be accurately planar and perpendicular to the specimen axis. Small deviations may lead to under-prediction of strength for the full range of rock strengths.
- Rock strength anisotropy requires careful consideration. The use of arbitrary test results for correlation between $Is_{(50)}$ and σ_c can easily lead to errors in k of 100%.
- Coring of soft rock may lead to poor recovery. Similarly, fractured rock may result in availability of small specimens for testing only. The point load strength test requires a smaller sample than a uni-axial compressive strength test and this can lead to bias, where uni-axial compressive strength testing takes place on the most competent specimens that are available and point load strength testing on less competent sections. This results in over-prediction of k .
- Scale effects for correlation between $Is_{(50)}$ and σ_c can result in under-prediction of k , where small-scale discontinuities control uni-axial compressive strength but not point load strength.
- Specimen water content can have significant impact on rock strength, with the strength increasing with decreasing water content. Point load strength testing in the field commonly considers "as-sampled" water contents only. Specimens for uni-axial compressive strength testing may have lower water contents because of drying during sample handling and transport. It is also possible, that laboratory practice considers specimen (re-)saturation by immersion.
- Small statistical populations may require exclusion of extreme values from the correlation, in particular for situations with significant rock heterogeneity.

References:

Abbs, A.F. (1985), "The Use of the Point Load Index in Weak Carbonate Rocks", ASTM STP 883 Strength Testing of Marine Sediments: Laboratory and In-Situ Measurements, R.C. Chaney and K.R. Demars, Eds, pp. 413-421.

ISRM (1985), "Suggested Methods for Determining Point Load Strength", Int. Jnl. Rock Mech. Min. Sci. and Geomechanical Abstr., Vol. 22, No. 2, pp. 51-60

APPENDIX 12 COMPRESSION WAVE VELOCITY

Typical values of compression wave velocity are given below.

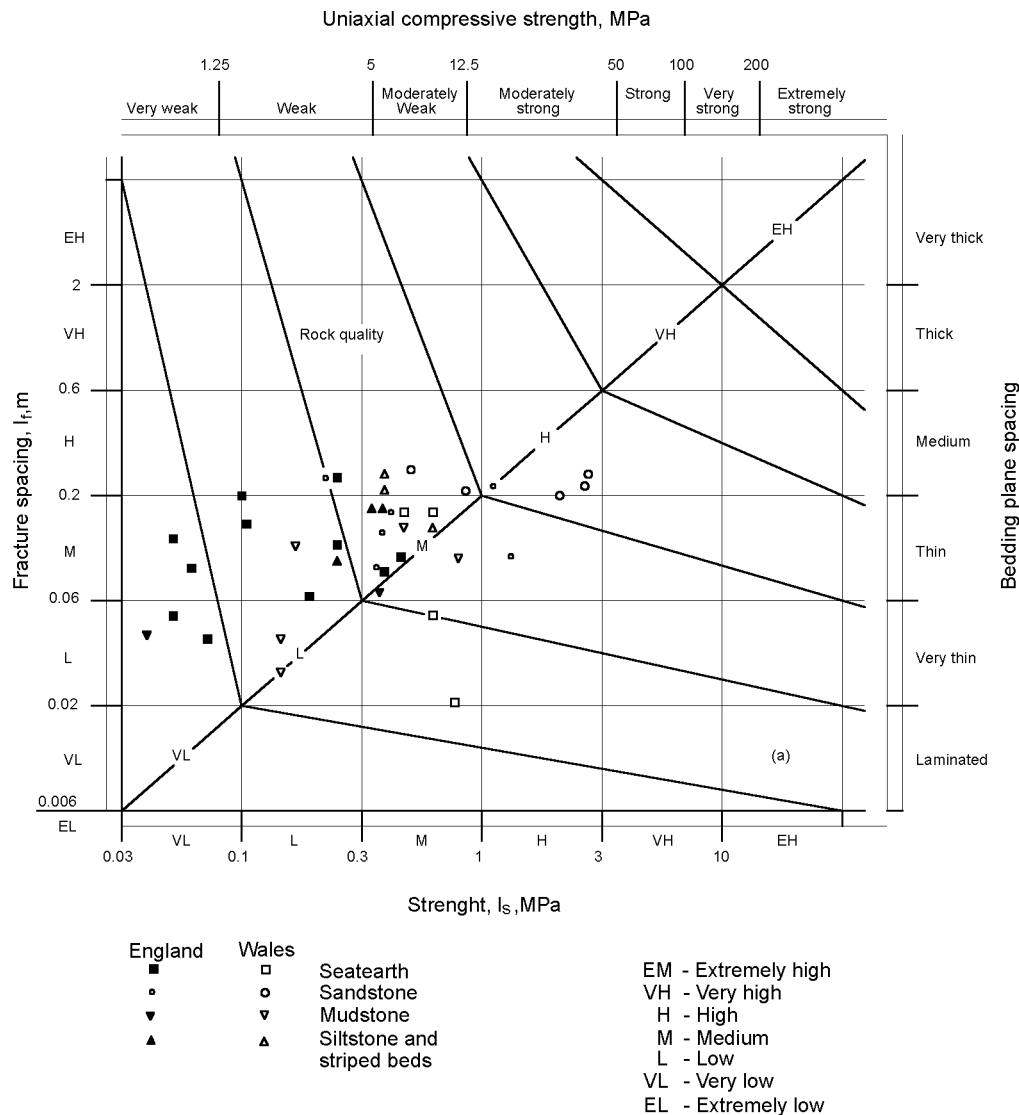
Ground Type	Compression Wave Velocity v_p (m/s)
weathered surface material	200 to 600
gravel or dry sand	450 to 900
sand (saturated)	1200 to 1800
clay (saturated)	900 to 2800
water*	1400 to 1700
sea water*	1400 to 1500
sandstone	1800 to 4000
shale	2700 to 4300
chalk	1800 to 4000
limestone	2100 to 6100
granite	4500 to 5800
metamorphic rock	3000 to 7000

* depending on temperature and salt content

Note: adapted from ASTM D5777

APPENDIX 13 ROCK MASS QUALITY AND ROCK EXCAVATION

Prediction methods for design of rock cuttings and rock excavation production rates generally follow some form of "rock mass rating" system, with the more complex methods requiring a wide range of input parameters. A general purpose qualification of rock mass quality, based on work by the Geological Society of London, is shown in Figure A13.1. It is one of a number of methods available.



I_s = point load index

I_f = fracture spacing index

Figure A13.1 Rock Mass Quality Classification Diagram

For ease of excavation, most of the methods consider the feasibility of a particular method. Production estimates are usually qualitative only. Braybrooke (1988) and MacGregor et al. (1994) present summaries. Figure A13.2 presents an example method for rock excavation prediction, as adapted from Braybrooke (1988).

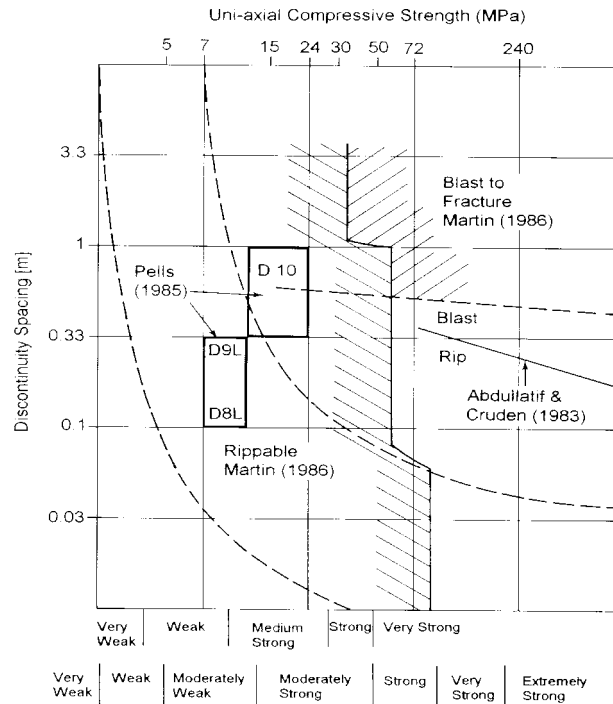


Figure A13.2 Example of Rock Mass Rating for Rock Excavation

Notes and references:

- The upper rock strength description is according to the original diagram and the lower description is according to BS 5930.
- Braybrooke J.C. (1988), "The State of the Art of Rock Cuttability and Rippability Prediction", Proc. 5th Australia-New Zealand Conf. on Geomechanics, Sydney, pp. 13-42.
- MacGregor, F., Fell, R., Mostyn, G.R., Hocking, G. and McNally, G. (1994), "The Estimation of Rock Rippability", Quarterly Journal of Engineering Geology, Vol. 27, pp. 123-144.
- Pells P.J.N. (1985), "Engineering Properties of the Hawkesbury Sandstone", in Engineering Geology of the Sydney Region, Pells P.J.N. ed., pp. 179-197.
- Martin J.M. (1986), "Predicting the Rippability of Sandstone in SE Queensland", 13th ARRB/5th REAAA Conf., pp. 119-132.
- Abdullatif O.M. and Cruden D.M. (1983), "The Relationship between Rock Mass Quality and Ease of Excavation", Bull. Int. Assoc. Eng. Geol., No. 28, pp. 183-187.

APPENDIX 14 SOIL PERMEABILITY

The following diagram presents typical values for soil permeability, as adapted from Carter and Bentley (1991) and BS 8004.

Coefficient of permeability	{m/s}	10 ⁻⁹	10 ⁻⁸	10 ⁻⁷	10 ⁻⁶	10 ⁻⁵	10 ⁻⁴	10 ⁻³	10 ⁻²
Permeability	Practically impermeable	Very Low		Low		Medium		High	
Drainage conditions	Practically impermeable		Poor		Good				
Typical soil groups	GC→ CH	GM→ SC MH MC-CL	SM		SW→ SP→		GW→ GP→		
Soil types	Homogeneous clays below the zone of weathering	Silts, fine sands, silty sands, glacial till, stratified clays				Clean sands, sand and gravel mixtures			Clean gravels
		Fissured and weathered clays and clays modified by the effects of vegetation							

Note: Carter M. and Bentley S.P. (1991), "Correlations of Soil Properties", Pentech Press Limited, London, 130 pp.

APPENDIX 15 ROTARY CORE BARRELS

SIZES OF ROTARY CORE BARRELS			
Designation	Core barrel design	Nominal diameter of core (mm)	Nominal diameter of hole, (mm)
B	BWF, BWG or BWM	42.0	60.0
B	BWT	44.5	60.0
N	NFW, NWG or NWM	54.5	76.0
N	NWT	58.7	76.0
NQ3		45.0	76.0
NQ		47.5	76.0
H	HWF or HWG	76.0	99.0
H	HWT	80.9	99.0
HQ		61.0	99.0
P	PWF	92.0	121.0
PQ		83.0	121.0
S	SWF	112.5	146.0
TBX		45.0	60.0
TNX		61.0	76.0
U	UWF	140.0	175.0
Z	ZWF	165.0	200.0
NMLC		52.0	76.0
HMLC		63.5	99.0
T2 56	T2 56	42.0	56.0
TT 56	TT 56	45.5	56.0
T6 66	T6 66	47.0	66.0
T2 66	T2 66	52.0	66.0
T6 76	T6 76	57.0	76.0
T2 76	T2 76	62.0	76.0
T6 86	T6 86	67.0	86.0
T2 86	T2 86	72.0	86.0
T6 101	T6 101	79.0	101.0
T2 101	T2 101	84.0	101.0
T6 116	T6 116	93.0	116.0
T6 131	T6 131	108.0	131.0
SK6 146	SK6 146	102.0	146.0
T6 146	T6 146	123.0	146.0

Notes (Adapted from BS5930 and ASTM D2113):

1. TBX and TNX are thin wall
2. NQ3 is triple tube wireline (also called double tube with retrievable inner tube)
3. Q is wireline
4. NMLC and HMLC are triple tube
5. TT is extra thin wall water barrel
6. T2 is water barrel
7. T6 is mud/water
8. SK is wireline barrel air/water/mud

APPENDIX 16 GEODETIC TERMS

General

- Geoid: equipotential surface coinciding with mean sea level in the oceans and its fictitious continuation on land.
- Spheroid: ellipsoid approximation of the earth's shape.
- Latitude: angle between the spheroidal normal through the reference location point, and the plane of the rotated major axis of the spheroid (equator).
- Longitude: angle between the meridian passing through Greenwich, and the meridian upon which the reference location point is lying.
- Parallel: line of equal latitude.
- Meridian: line of equal longitude.
- Central Meridian: line where the spheroid touches the cylinder for a conformal cylindrical projection.
- Projection: systematic representation of a portion of the earth's surface upon a plane.
- Universal Transverse Mercator (UTM): special case of a conformal cylindrical projection.
- Geodetic Datum: datum fixed by orientation and position of the spheroid.

Location control terms

- Location Reference Point: point to be surveyed, such as borehole axis for positioning survey and/or ground surface for elevation survey.
- Ground Surface: top of natural or man-made ground, either above or below the water level, at a particular date and time.
- Water Depth: vertical distance between water level and ground surface at a particular date and time.
- Water Level: elevation of free water surface at a particular date and time.
- Grid: rectangular plane co-ordinate system.
- Network: geometric net of control points.
- Baseline: line between two control points.
- Control Point: location for which the co-ordinates and/or elevation are known.
- Monument: fixed control point of network.
- Bench Mark: monument used as reference for vertical control and/or horizontal control.
- Elevation: vertical distance above or below datum.
- Datum: fixed reference elevation, usually related to LAT or mean sea level.
- Chart Datum: datum applicable to a hydrographic, bathymetric or navigation chart.
- HAT: Highest Astronomical Tide.
- MHHW: Mean Highest High Water.
- MHW: Mean High Water.
- MSL: Mean Sea Level.
- MLW: Mean Low Water.
- MLLW: Mean Lowest Low Water.
- LAT: Lowest Astronomical Tide.

Survey method terms

Global Positioning System (GPS)

- GPS is a globally used system which uses satellite signals to give position, often with accuracy improved by calibration at a local base station.
- Small hand held units are available which currently give a typical accuracy of from 2 m to 5 m provided they can be calibrated against a local base station. This is useful for walkover and scouting surveys (accuracy without a base station is generally inadequate at 2 m to 300 m).
- Differential Global Positioning System uses at least 2 receivers at the same time, and is a more accurate version of GPS. It is generally accurate enough for the setting out of site investigation works.
- GPS and DGPS systems are continually becoming smaller, more accurate and more portable with better electronics and improved satellite cover.

Ground Positioning

- Direction Method: angle measurement system for horizontal control whereby the direction of a line is given by the angle it forms with a baseline.
- Angle Method: angle measurement system for horizontal control whereby individual angles between neighbouring lines are measured independently in a number of sets.
- Intersection Distance Method: distance measurement system for horizontal control whereby distances to a location are measured from two control points.
- Extension Method: distance measurement system for horizontal control whereby distances are measured from extension of a location on a baseline to the intersection with another baseline.
- Intersection Angle Method: angle measurement system for horizontal control whereby angles to a location are measured from two control points on a baseline.
- Orthogonal Method: distance and angle measurement system for horizontal control whereby distances to a location are measured along and perpendicular to a baseline.
- Polar Method: distance and angle measurement system for horizontal control whereby distance and angle to a location are measured relative to a control point on a baseline.

Elevation Survey

- Differential Levelling: elevation survey method using levelling instrument for vertical control.
- Trigonometric Levelling: elevation survey method using vertical-angle measurements.
- Water Level Reduction: elevation survey method using height measurements relative to water level.

Water Depth Survey

- Direct Sounding: water depth survey method using mechanical distance devices such as hand lead, depth pole, drill pipe.
- Pressure Sensing: water depth survey method using water pressure device.
- Echo sounding: water depth survey method using echo sounder.

Ground positioning apparatus

- Tacheometer: special theodolite including optical distance measurement.
- Electromagnetic Distance Measurement (EDM): distance measurement system comprising a modulated signal transmitted from one end of the measured distance and reflected or retransmitted back at the other end.
- Theodolite: instrument for angle measurement.
- Prism: instrument for fixed angle measurement.

- Invar Subtense Bar: device used for optical determination of distance.
- Compass: instrument providing angle relative to magnetic North.
- Gyro Compass: instrument providing angle relative to absolute North as dictated by meridians.

Elevation survey apparatus

- Levelling Instrument: optical device using horizontal sighting for vertical distance measurement.
- Levelling Staff: graduated length measurement rod used for optical elevation survey.
- Circular Level: instrument used to check verticality.
- Surveying Tape: mechanical distance measurement instrument.

In addition, elevation survey apparatus can include the tacheometer, the theodolite and the invar subtense bar.

Water depth survey apparatus

- Hand Lead: direct method using marked-wire or surveying tape measurements.
- Depth Pole: direct method using pole measurements.
- Water Pressure Device: apparatus for determining height of water based on pressure measurements relative to a reference pressure.
- Differential Water Pressure Device: apparatus for determining height of water based on pressure measurements relative to atmospheric pressure or hydrostatic pressure.
- Echo sounder: acoustic distance measurement apparatus using a fixed beam with a vertical axis and measuring the range by timing the two-way journey of a pulse of acoustic energy.
- Bathymeter: device combining water pressure and echo sounder measurements.
- Motion Compensator: device for correction of echo sounder measurements to compensate for heave, pitch and roll of the echo sounder.
- Tide Gauge: fixed depth pole used for measurement of periodic variations in water level.